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SAMINT-MILI-26012
Master's Thesis 30 credits
May 2026

Benchmarking AI Tool Adoption in Cost Engineering

An Integrated TOE Framework for AI Tool Assessment and Prioritized Recommendations for an Automotive OEM Group

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Master's Programmed in Industrial Management and Innovation

Master program i industriell ledning och innovation

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Abstract

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The automotive industry operates under significant cost pressure driven by global competition, supply chain disruptions, and shifting demand patterns, making the cost engineering function a critical interface between procurement and Research & Development (R&D). While Artificial Intelligence (AI) tools show strong potential to address the traditionally manual and resource-intensive nature of the cost estimation process, automotive Original Equipment Manufacturers (OEMs) face challenges due to a fragmented landscape of available solutions, often investing in AI tools that fail to sustain their intended value over time. This thesis investigates this issue through a structured evaluation of AI tools within the cost engineering function at Company X, an automotive OEM, and its four peer organizations (A, B, C, and D), all of which are undergoing active digital transformation.

The study applies an integrated Technology-Organization-Environment (TOE) benchmark framework developed for comparative tool ranking and assessment to support adoption recommendations. Six measurable factors – Technical Scope, Integration, Use Cases & Value, Workflow & Culture, External Drivers, and Ecosystem Readiness – are scored across four maturity levels from L1 to L4. The research design follows a qualitative study with semi-structured interviews, a systematic literature review, and an internal observational and document study. Interview studies 1 and 2 were conducted with the Company X cost engineering team and the IT head. Triangulation of these interviews, the literature review, and management requirements at Company X yielded the six factors and their maturity-level definitions within the TOE dimensions. Interview study 3 was conducted with AI tool developers and users to capture the benchmark framework outcome across ten AI tools from Company X and four peer organizations (A, B, C, and D).

The results shows that although Company X has started initiatives towards implementing AI-driven solutions, several critical areas still need to be addressed before efficiency and improvement targets can be reached. The benchmark scores ten AI tools and provides a practical recommendation order for adoption in the cost engineering function at Company X. The benchmark reveals three patterns across the studied OEM group: a shared pilot ceiling stage, data-tool mismatch where the tools score low on Ecosystem Readiness, and a strategic move as the driver for adoption. To adopt AI tools successfully, Company X must strengthen its readiness across the six factors and follow a prioritized recommendation matrix for the ten AI tools currently available. Despite the challenges, Company X has the necessary foundation for AI-driven transformation, provided the preconditions are addressed.

Keywords: Cost engineering, Automotive OEM, Artificial Intelligence Tools, TOE Framework, Benchmarking, AI adoption, Cost estimation, Case study

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SAMINT-MILI-26012
Printed by: Uppsala Universitet

Popular Scientific Summary

Businesses across industries face constant pressure to reduce costs while maintaining product quality and customer satisfaction. In the automotive sector, this pressure is especially intense due to strong global competition, rising material costs, and the challenge of sourcing thousands of parts from suppliers worldwide while keeping prices competitive. This work is assigned to cost engineers, who are part of the procurement function and are responsible for analyzing part costs and using that knowledge to negotiate more effectively with suppliers. Today, most of this work is still done manually, which is time-consuming, error-prone, and heavily dependent on individual expertise, making it difficult to scale across teams and companies.

Artificial Intelligence (AI) tools that can automatically read supplier documents, compare and predict part prices, and analyze cost patterns are now widely available, and many automotive manufacturers have started testing them. But a recurring pattern emerges: companies invest in AI tools, run promising pilots, and then watch those tools lose their purpose before they reach everyday use. This research set out to understand why, using automotive Company X and its four peer organizations as a real-world case study. Interviews were conducted with the cost engineers and the Information Technology (IT) head of Company X, and with users of ten AI tools currently used or piloted across Company X and the peer organizations. The evidence was analyzed using an integrated Technology, Organization and Environment (TOE) benchmark framework developed specifically for this case study, which scores AI tools across six factors and produces both a comparative ranking and a prioritized set of recommendations for adoption by Company X.

The findings show the broad picture: the obstacle is not the technology alone, but the level of development of the tool and the organization trying to adopt it. Three consistent challenges were found: data fragmentation across systems, workflows, and supplier data formats. The benchmark and ranking produced in this study provide a clear picture of where each company stands, identify absorption deficits, and recommend specific actions for Company X as it seeks to acquire the new technology. The integrated TOE benchmark framework developed here is a transferable structure that other industrial organizations facing similar AI adoption decisions, particularly those still in the pilot stage or preparing to invest in AI for cost engineering and procurement, can adapt as a starting point for their own context by revisiting the factor triangulation, the reverse-scoring direction, and the benchmark re-scoring for their own portfolio.

Acknowledgement

This thesis marks the final project of our master's studies at Uppsala University, Sweden. We would like to express our sincere thanks to our supervisor and the head of the department at Company X for their trust and for this opportunity. Special thanks to the whole cost engineering team at Company X and other interviewees for their openness and willingness to engage. Their support helped structure the work, provided guidance, and feedback throughout the process. We are grateful to our subject reader, Morteza Ghobakhloo, for his enthusiasm, constructive critique, and valuable feedback. We would also like to extend our gratitude to the examiner, Michał Zawadzki, and faculty members of the Department of Civil and Industrial Engineering at Uppsala University. Finally, we owe a heartfelt thanks to our family and friends for their constant support throughout this journey.

The thesis work was evenly distributed, and all sections were written entirely by the authors. We collaborated closely, with continuous proofreading and collective revisions.

Keshav Pandiyan & Surendar Selvaraj
Uppsala, May 2026

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List of Abbreviations

Abbreviations	Expansion
AI	Artificial Intelligence
R&D	Research and Development
OEMs	Original Equipment Manufacturers
ERP	Enterprise Resource Planning
CBD	Cost Break Down
BPR	Business Process Re-Engineering
RFQ	Request for Quotation
TOE	Technology-Organization-Environment
IT	Information Technology
AIMM	Artificial Intelligence Maturity Model
AI-CAM	Artificial Intelligence Capability Assessment Model
CMM	Capability Maturity Model
CRISP-DM	Cross-Industry Standard Process for Data Mining
ML	Machine Learning
NN	Neural Network
NLP	Natural Language Processing
LLM	Large Language Model
GenAI	Generative Artificial Intelligence
GDPR	General Data Protection Regulation
SMEs	Small and Medium-sized Enterprises
API	Application Programming Interface
PLM	Product Lifecycle Management
SRM	Supplier Relationship Management
CoE	Centre of Excellence
ESG	Environmental, Social, and Governance
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
JSON	JavaScript Object Notation
VOS	Visualization of Similarities
MES	Manufacturing Execution System
CAD	Computer-Aided Design
PCB	Printed Circuit Board
B2B	Business-to-Business
RQ1	Research Question One
RQ2	Research Question Two
RQ3	Research Question Three

1. Introduction

The purpose of this chapter is to give a broader overview of the cost pressure that industries face and the need to optimize cost through automation and AI, together with the role of the cost engineering function in this context. The study focuses on Company X to explore the case and detailed information related to the background, problem statement, aim, purpose, and research questions.

1.1 Background

Industrial companies are operating under cost pressure that rarely eases in current market conditions. A part of it is the classic competition where rivals improve performance and features while pushing price and lead-time expectations downward. However, the newer layer of pressure comes from geopolitical tensions, supply chain disruptions, and shifting demand patterns, which are the key reasons forcing firms to replan faster, making cost control harder and more expensive. In such an environment, cost reduction is no longer a one-time savings program but a continuous capability to spot cost drivers early, respond to disruptions, and achieve internal cost targets. That is the reason many firms are accelerating digitalization and automation across core functions.

One such sector where this becomes clearly visible is automotive and industrial manufacturing, where Original Equipment Manufacturers (OEMs) operate under global supply networks, platform complexity, and high investment, making cost a central management discipline. For example, Volvo Group describes 2024 as a year of “normalizing market demand alongside continuous geopolitical turmoil” (Volvo Group, 2024), highlighting that resilience and cost are tested even when volumes soften.

In automotive procurement, the storyline is explicit that cost, supply assurance, and resilience are to be managed together. The procurement focus includes safeguarding the supply of vehicle parts and optimizing cost, which is directly linked to the global disruptions experienced after 2020 (Volkswagen Group, 2024). As companies push this agenda forward, the focus naturally narrows from industry-wide cost pressure to the functions that can move the needle. Procurement is one of them, because it directly influences cost through supplier selection, contract terms, and category strategies. However, procurement performance increasingly depends on what happens upstream, specifically, the quality of early cost logic, target setting, and should-costing for industry-wide negotiation strategies and sourcing choices. This is where the cost engineering function becomes critical and supports early project decisions when design freedom is high, information is incomplete, and wrong assumptions get locked into the product. Cost engineering refers to the discipline of estimating, analyzing, and controlling the cost of products throughout the development and procurement lifecycle, with a combination of engineering knowledge of materials, processes, and design, and a commercial point of view for fair negotiations. In the automotive sector, cost engineering operates at the interface between R&D and procurement, providing a

foundation for cost estimation and supplier negotiation. Consequently, if the cost engineering function is slow or inconsistent, procurement either negotiates without a strong anchor or inherits expensive design decisions that are difficult to reverse at later stages of development.

Traditional cost engineering approaches struggle to keep up with today's complex and data-intensive environment (Bodendorf, Merkl and Franke, 2021, p.2). Cost inputs are scattered across engineering tools, Enterprise Resource Planning (ERP) systems, procurement platforms, supplier quotes, and legacy spreadsheets, while product variety and supply risk grow alongside shrinking decision cycles. This mismatch is a major reason AI is increasingly discussed as decision support rather than a standalone automation project. Research on AI in cost engineering emphasizes both the potential of AI and the adoption frictions, namely data quality, governance, and integration into real work (Guida et al., 2023). Similarly, the work on automation in procurement shows that efficiency gains come with organizational and coordination implications that must be managed (Viale and Zouari, 2020; Bodendorf and Franke, 2024, p.3).

Consequently, attention increasingly shifts toward industries facing sustained cost pressure driven by competition, market volatility, and internal cost targets. Companies respond through the digitalization of procurement because it is a direct cost factor and a resilience function. Subsequently, attention moves upstream into sourcing strategy and cost engineering, because early decisions determine a large share of lifecycle cost. As the complexity rises, organizations look towards AI-based decision support to improve speed and consistency, but the practical challenge becomes how to embed it into workflows and governance rather than treating it as a plug-in tool (Wang, 2025, p.1).

1.2 Problem Statement

Across the automotive industry, procurement and cost engineering are undergoing an active transition from traditional approaches towards modern data-supported and AI-supported methods. While leading OEMs have begun to integrate AI into their procurement and supply chain processes, many organizations still face challenges in adopting these technologies into practical implementation within their cost engineering workflows.

One such company in the transformation phase is Company X. From an industrial perspective, AI interest is publicly visible in other automotive OEMs. Volkswagen Group explicitly frames "digitalized supply chains and automated processes as part of its transformation and describes AI use cases tied to monetary and quality-related cost savings" (Volkswagen Group, 2024). Meanwhile, BMW Group (2025) highlights the use of Generative Artificial Intelligence (GenAI) in purchasing to improve efficiency in information work and decision-making within its supplier network. To narrow down and look within the cost engineering function at Company X, understanding the current way of working is needed to support the transition. A pre-study was done by Company X to find out how the cost estimation activities are organized, how different departments interact, and how the overall

process is executed in practice, from new product cost initiation to final cost settlement with the supplier.

The findings are based on observation and interview studies to point out the manual processes carried out by the cost engineering team and the initial proposal of areas where automation / AI can be integrated. Further, it maps the organizational readiness at Company X to accommodate these changes. According to Bhogal and Rudqvist (2025), the cost engineering function operates in close collaboration with both R&D and procurement departments. R&D is responsible for developing product designs, technical specifications, and modern technologies, while procurement ensures that suppliers, materials, and components meet the required cost, quality, delivery, and sustainability targets.

The operational workflow at Company X follows a structured but manually intensive process: a purchaser submits a support request that contains the article number, project description, estimated volumes, technical drawings, specifications, deadlines, and supplier-provided Cost Break Downs (CBDs). Then the cost engineer's role starts with a should-cost analysis using internal legacy systems and Excel software. Finally, the results are validated against supplier quotations and shared back with the purchaser for negotiation. A detailed visual representation of the cost engineering workflow at Company X, including the handover process, cost estimation steps, and validation procedures, is documented in Bhogal and Rudqvist (2025) from figures 8 to 10. The present study builds on this process understanding, which was verified using an observational study described in section 4.4.3.

Building on the workflow described above and on the pre-study by Bhogal and Rudqvist (2025), six gaps were identified where automation and AI can be integrated: inefficient process initiation; manual and time-consuming data gathering; challenges in data processing and storage; limitations in the cost calculation tool; inefficient validation and supplier comparison; and the late involvement of cost engineers. The previous study covered the first four steps of the Business Process Re-Engineering (BPR) framework, identifying areas for improvement in the existing workflow. The remaining two steps, namely, what new processes need to be implemented and how they can be assessed in practice at Company X, constitute the gap this thesis addresses.

The second part of a previous study conducted by Bhogal and Rudqvist (2025) gives insights into the readiness assessment of Company X at the organizational level. The dimensions analyzed are data quality, data infrastructure, education, change management, organizational adaptability, strong leadership, and cross-functional collaboration. Finally, it is rated using a readiness assessment, which shows that education is at a mature level. Data quality, strong leadership, and cross-functional collaboration are at a developing level. Data infrastructure, change management, and organizational adaptability are still at an emerging level.

1.3 Aim and Purpose

This thesis aims to explore the AI tools that can benefit the cost engineering function of Company X by benchmarking current practices across the peer organizations. The output is an integrated TOE benchmark framework grounded in process improvement, together with a practical recommendation for adoption. The purpose is to focus on existing or developing AI tools that are within the scope of the peer organizations, which increases the chance of acceptance within Company X.

In addition, learning the basic AI concepts is essential for team members before introducing new tools into the existing workspace. AI should be understood as a way for procurement and cost engineering teams to extract and use information more consistently, not as a replacement for the judgment or experience of a professional. The thematic focus of a systematic literature review on AI in procurement indicates that the primary contribution of AI lies in automating the extraction of information and facilitating multi-criteria decision-making within purchasing environments (Balkan and Akyuz, 2025).

Three major categories of data are relevant to AI in cost engineering and procurement. The first is structured transactional data, such as purchase orders and supplier master records. The second type of data consists of semi-structured/unstructured text, i.e., Requests for Quotation (RFQs), offers, contracts, and requirement descriptions. Lastly, relational/networked data provides value when understanding the relationships between suppliers, parts, and tiers, not just within record databases. As an example, graph-based learning has been used to infer previously unknown connections between suppliers, helping organizations to manage their supplier networks more effectively and to enhance awareness of potential risks in complex supply chains (Kosasih and Brintrup, 2022). Explainable AI is critical when AI impacts decisions with commercial or operational results. According to a study by Modaresi (2024), in the context of explainable AI for supply chain risk classification, stakeholders must understand the model's output to trust and act on it. This is particularly important as Company X develops its AI tools, because the technical work must also include some form of defensibility so that its stakeholders can use the model's output appropriately.

Finally, these tools will only be useful at scale when organizations also build the right infrastructure (in terms of data readiness, integration, governance, and analytical capabilities) to support “big data” initiatives. As emphasized by Contreras Pinochet et al., (2021), the benefit of big data analytics capability does not stem from the volume of data alone but from how successful firms use the data to inform operational decisions.

After understanding what is useful in cost engineering practices that need to be changed, the next direction is how the decision is to be made at the organizational level about the proceeding with the insights. Integrating the best practices and recommendations for AI adoption, a decision framework is developed from actionable outcomes for practical recommendations. This helps Company X stay updated within its peer organizations and be in a competitive position with the latest technology in the market.

1.4 Research Questions

Research Question One (RQ1) – What AI tools and use cases are present in cost engineering and procurement across Company X and the peer automotive OEM group, and what organizational and environmental conditions do they show in practice?

Research Question Two (RQ2) – How can AI tools used across Company X and its peer organizations be benchmarked using integrated TOE dimensions?

Research Question Three (RQ3) – How can the integrated TOE benchmark insights be developed into prioritized AI adoption decisions for Company X's cost engineering function?

The three research questions are addressed through the methods described in Chapter 4. RQ1 is addressed by mapping the AI tools and their use cases across Company X and the peer organizations, drawing on the systematic literature review and the semi-structured interviews. RQ2 is addressed by benchmarking these AI tools on the six factors of the integrated TOE framework, where the interview evidence is coded through thematic analysis and scored to produce a comparative ranking. RQ3 is addressed by reading the same benchmark scores as readiness signals and combining them with the earlier functional-level study at Company X (Bhogal and Rudqvist, 2025) to arrive at prioritized recommendations for AI tool adoption in the cost engineering function.

2. Literature Review

This chapter establishes a conceptual and empirical foundation for the study. The aim of this chapter is to understand and analyze the cost engineering and procurement function from an industrial perspective, its traditional approach that is widely used, the limitations appearing in current conditions, the improvements that can be made using digitalization with AI tools, and the path to find a suitable option to work from the organization's point of view. Its purpose is not simply to survey what has been written about AI in procurement and cost engineering, but to build a reasoned argument for why the research questions in this thesis are necessary and how existing knowledge motivates the need for this study. The chapter begins by locating cost engineering within the broader procurement context and explaining why cost estimation carries strategic importance in industrial organizations. It then examines the functioning and limitations of traditional cost engineering methods, before discussing the role of AI and digitalization as decision-support tools in procurement and cost engineering. This leads to the central analytical insight that organizations face both a selection problem, where identifying suitable AI tools is complex, and a readiness problem, where successful adoption depends on organizational and contextual factors. All these topics are discussed from various perspectives, drawing on existing research as the reference base throughout. In this way, the chapter directly supports the research questions and provides the analytical foundation for the framework developed later in the study.

2.1 Cost Engineering and Procurement in Industrial Organization

Cost engineering is a function within the procurement team that becomes involved in new product development activities once a project is initiated, and modern design specifications are in place across industrial organizations. Most complex products pass through this procurement process, making procurement a dominant share of total expenditure for the organization. Research indicates that "between 50 and 70 percent of a company's total expenditure is spent on procurement" (Berggren and Matti, 2021, p.1), making procurement a cost control center for maintaining competitiveness. This highlights that cost engineering is not only an operational function, but also a key mechanism through which organizations manage cost performance and maintain their competitiveness in complex industrial environments.

Within this context, estimating the cost of a new product plays a critical role in the decision-making process; it supports supplier negotiations, pricing strategies, and early-stage development activity. In manufacturing industries, particularly automotive, early cost visibility is essential due to the high proportion of direct material cost in the total product structure (Hammann, 2024, p.2). This highlights the primary role of cost engineering as a support function bridging R&D and procurement teams, translating technical and commercial inputs into analytical cost estimates. It provides real-time tentative cost information compared to the target cost and the supplier's request. These estimates are often used to anchor the suppliers' negotiations and decisions in sourcing. At the same time, the reliance

on multiple data sources and cross-functional inputs makes the estimation process complex and dependent on both data quality and expert judgment.

Having these insights guides the procurement team in each organization toward a strategic approach to negotiating with suppliers to fix the actual cost, which is usually the expenditure to the company, and determines the profit return. To analyze the cost structure, the cost engineering team should also understand the manufacturing processes, supplier capabilities, and risk factors that lead to the elimination of "black box" items, which make the sourcing stage transparent for both sides. Understanding this role is important because the challenges that AI aims to address in cost engineering are not abstract technical issues but practical limitations in how these functions currently operate within industrial organizations.

2.2 Traditional Cost Engineering Approaches and Their Limitations

The traditional approach to cost estimation involves the following activities: collecting data, verifying cost information against knowledge databases, applying standard inputs and processes for the products, and final verification against the target cost. These approaches can be broadly categorized into two techniques: qualitative techniques are exclusively based on expert judgments or analogical reasoning, and quantitative techniques, consisting of parametric and analytical models. In short, cost engineering relies on methods such as parametric estimation, analytical cost estimation, and expert-based approaches. Hammann (2024) states that these methods use historical data, engineering relationships, and domain expertise to estimate product costs and support procurement decisions. Parametric models identify the statistical relationships between the product features and cost, while analytical approaches break down products into individual components and processes to calculate cost from the bottom up. Traditionally, cost engineering methods have relied heavily on expert judgment during the early phases of product development and less so toward the end of development, due to the limited information available at earlier stages. For this reason, practical information is limited, and expert judgment and analogical reasoning are often used to produce initial estimates. These methods offer a structured and comprehensible foundation for cost estimation, which is crucial for transparency in supplier negotiations and in procurement.

Despite the variations in tools and internal systems, the fundamental approach to cost engineering remains highly similar across organizations and in automotive. This knowledge is often developed through experience, making it difficult to fully code or automate. Many organizations continue to rely on established tools such as spreadsheets and ERP systems, which makes them reliable and transparent to management for cost decisions since they have direct financial implications. This creates a preference for deterministic and rule-based models over complex alternatives (Manuguerra et al., 2023, p.2). There are technical and organizational reasons for this preference, but the ability to provide clearly defined and auditable cost structures by these methods serves as a basis for negotiations with suppliers and justification for making internal decisions. Also, it provides explainable and justifiable

workings, which can be a tool for negotiations with suppliers and for the future scope of business.

However, the same characteristics that make traditional cost engineering methods reliable and transparent are also the source of their greatest limitations in today's industrial setting, characterized by high product complexity and rapid changes in the marketplace. A key challenge is cost estimation, often performed in the early stages with limited information; expert knowledge can sometimes lead to uncertainty in decision-making due to assumption mismatches with actual cost conditions. “Overestimation of cost could mistakenly discard profitable products, while underestimation creates the risk of producing at a loss” (Hammann, 2024, p.2). The risk is always associated with the product, since the decision about so many variables is made by a single human's knowledge and experience. This highlights a structural limitation where the quality of all decisions made in these environments relies heavily on the expertise of the individual making the decision rather than on systematic and scalable methods for evaluating costs. Cost estimation by human expertise alone is a double-sided trade-off, since decisions are made on attributes that are not fully known. This limitation in early-stage estimates is extremely sensitive and difficult to rely on, especially when design parameters are still evolving, and each change in the design phase will involve a lot of time in estimating the cost.

Furthermore, traditional approaches rely on historical data, which may be outdated or inconsistent. On the other hand, the information from supplier quotations limits proactive decision-making, as obtaining such data for multiple conditions is impossible. This dependency on both reduces the flexibility and slows down the evaluation process even more, particularly when a quick comparison is required. Additionally, traditional cost engineering methods were not intended to handle the volume, variety, and velocity of data being generated in today’s procurement environments. At the same time, data as a collective is incredibly huge in organizations that are being underutilized and difficult to process with human expertise. This finding is supported by Mirasçı and Aksoy (2025, p. 1), who state that the purchasing department generates a vast amount of data, but it is frequently not fully exploited. This indicates how the data is perceived, which leads to limitations in its use in effective cost estimation activities. Traditional methods are not well suited for large-scale processes, complex datasets, or for capturing the nonlinear relationships between cost drivers, which limit the ability to fully leverage available information for efficient cost calculations. As a result, the cost engineering process remains reactive, dependent on manual analysis and expert judgment, constrained in its ability to support fast, data-driven decisions. The advancement of AI for cost engineering has been made possible by several factors: the early stage of uncertainty, the restricted application of available data, and the increasing complexity of sourcing decisions. However, evaluating whether AI is a practical solution requires more than acknowledging that it exists; it also means examining its performance to see whether there are true barriers to its success.

2.3 AI Tools, and Decision Complexity in Cost Engineering

The limitations of traditional methods have led to the increasing adoption of digitalization and AI in cost engineering, procurement, and design functions. AI-based models can process large datasets, identify complex cost relationships, and improve prediction accuracy. Research shows that Machine Learning (ML) can support cost estimation by analyzing historical data and identifying patterns between product features and costs (Bodendorf, Merkl and Franke, 2021, p.3). The predictive power of AI is an important advantage in the early stages of product development, where actual design data may not yet be available even though financial commitments are already being made. At these early points, AI allows for the more rapid estimation of costs and decision-making processes that are more consistent and scalable than a purely human-based method would allow. It can improve consistency and performance time by leveraging available data capabilities to provide more informed decision support. This predictive nature of cost estimation gives the upper hand in negotiations, enhancing cost transparency and support for effective resource allocation.

In practice, this transformation is supported by integrating AI models into existing digital infrastructures, such as ERP systems and procurement platforms, where they can continuously analyze incoming data, generate insights, and support decision-making across various stages of the cost engineering lifecycle. The literature emphasizes that AI should support rather than completely replace human expertise, acting as a decision-support tool within existing workflows (Bodendorf and Franke, 2024, p.2). In industrial environments, having a mixed-method approach to using AI is essential because organizations need to trust the decisions made through the AI process and have an auditable record of those decisions for purposes of managing their relationships with suppliers.

Beyond cost estimation, there are many other AI use cases that can support the organization, such as supplier evaluation, demand forecasting, contract analysis, and process automation (Balkan and Akyuz, 2025, p.2). While these use cases are outside the primary scope of this study, they form part of a broader procurement context. These various use cases of AI tools could incorporate multiple processes involved in procurement and may leverage diverse technology platforms such as Natural Language Processing (NLP), ML, Neural Network (NN) technology, and robotic automation.

However, while the potential of AI is well established in the literature, there is significant confusion associated with the many types of AI tools currently available, creating added complications for organizations seeking to implement them. While AI tools offer significant potential, it is the selection of the right tool that is widespread and confusing. Various studies highlight that tools and methods, including ML, NN, and automation solutions, operate across different areas of cost engineering and procurement. With the high number of processes involved, organizations have a wider range of options to choose from. AI use cases span multiple processes, from cost estimation and supplier selection to contract analysis and process automation (Balkan and Akyuz, 2025, p.2). This diversity creates complexity in the decision-making process to determine which AI tools are available in the market and which

capabilities the organization needs to benefit financially. There has also been little to no standardization in how the various AI tools are classified or evaluated across multiple organizations, as well as within studies conducted on AI.

Due to inconsistencies and a lack of standardization in the classification of applications, organizations face difficulties in selecting the right use of AI in cost engineering. Chotaliya (2024, p.6) makes the same point that several conflicting and overlapping concepts exist regarding opportunities to utilize AI in cost engineering. Even after knowing that AI will help in efficient cost engineering practices, organizations struggle to map the AI tools that can profit from their specific use cases. This indicates that the challenge is not only the absence of AI tools but also the absence of structured methods to validate and compare them. AI tools are dependent on factors such as product characteristics, data structure, supplier base, and internal processes, and these use cases will perform differently in each organization where they are employed.

Apart from technology and use cases, it is also important to understand the maturity level of the organization itself and of the technology it is considering. This will also assess the risk associated with using the new AI tool and how other organizations place it in their cost engineering function, whether it is fully developed or is still under development. Organizations also want to understand the performance impact before adopting a new tool. This will be a key area for problem-solving, comparing current cost engineering processes with the new AI tool and benchmarking them. However, AI maturity models or capability assessments are typically used to evaluate an organization's overall level of preparedness rather than determining which specific AI tools would best serve an organization's specific use case. Evaluating these factors provides a structured approach for the organization to understand where they stand with their current process, decision-making relating to new AI tools, and use cases specific to their context.

This reinforces the need for a more systematic and case-oriented approach where AI tools are assessed based on various factors that best suit a given setting. Each organization should analyze its own context; therefore, an integrated TOE benchmark framework that addresses case-specific requirements is necessary, as developed in Chapter 3. The key issue is not whether cost savings can be improved through AI, but how to evaluate and select the best application of AI to an organization's performance systematically and objectively.

2.4 Organizational Challenges in AI Adoption

With the technological aspect and its functionality level, the next crucial step is to understand the integration of new AI tools. Even if an organization identifies the appropriate AI tools for its cost engineering function, it does not automatically mean these tools will be implemented successfully, because many other factors will influence implementation; in particular, organizational and contextual factors play a decisive role. One barrier to the implementation of AI tools within industrial processes is how the AI tool integrates into and changes the existing workflow of the procurement function (Spreitzenbarth, Bode, and Stuckenschmidt,

2024, p.1). This integration into cost engineering is not just about modern technologies, but also the way that organizations execute their existing processes and how they will process and coordinate. Even though it is seen as a barrier for implementation, it gives the final direction to the organization to work and acts as an improvement point for the future roadmap.

Looking more closely at these barriers, one of the most critical aspects is data quality and availability: AI models depend on the volume of structured, reliable data, whereas the cost engineering team typically operates with fragmented and incomplete datasets. These types of datasets are often shared across multiple systems, such as procurement tools, supplier portals, ERP platforms, and spreadsheets, making integration complex. Integrating AI into such environments requires a significant redesign of data pipelines and IT infrastructure, making implementation complex and resource-intensive. This indicates that data readiness falls under the foundational system for AI adoption rather than being a secondary requirement. It depends solely on the specific organization and its requirements, which determine the need for this data structure and the technology to capture this fragmented information.

Another challenge is the organization itself, meaning the people working in the existing environment. Traditional cost engineering processes are labor-intensive and rule-based, limiting their ability to adapt to dynamic changes. This transition to AI-driven processes requires redesigning workflows, the way of thinking, and the use of modern technology, which introduces a lot of uncertainty and operational disruption for an initial period. The move from individual judgment-based decision-making to data-supported insights, enabled by the adoption of AI tools, also represents a significant transition; therefore, organizations will need to develop new competencies to ensure stakeholders trust the decision-making process that relies on AI. Resistance to change is a practical human behavior and remains a major barrier. Without proper training and change management, organizations struggle to embed AI into daily operations (Bhogal and Rudqvist, 2025, p.13). Employees might also take this change as a threat to their expertise and future positioning in the organization, leading to less adoption and trust between the team. In addition, the explainability steps up, as cost engineers must understand the work and justify AI outputs during negotiations with the supplier. Helping employees understand why the change is important, what benefits it brings, and how it can ease the work with available resources is therefore a critical point during the transition.

Alongside these internal factors, the external challenges affecting the use of AI in cost engineering and procurement also need to be considered. Since a large volume of data is involved, regulation, governance, and security influence the level of adoption and the limitations on usage at various levels within the organization. The organizations must ensure AI tools comply with internal regulations and standards when dealing with supplier-sensitive data, pricing, and contractual data. Thus, AI-based results should be transparent, auditable, and accountable, with explanations and justifications that can be provided during negotiations and protected within the context of organizational data privacy. Adoption decisions are also influenced by market and competitive forces, since companies must balance the difficulties of

early implementation against the risk of delayed adoption. These external constraints highlight the importance of a safe environment, which influences how and to what extent AI can have access to the information.

The literature confirms the analytical success of AI for cost estimation and procurement analysis; however, further research is still needed to understand how AI can be integrated into organizational decision-making processes within real industrial settings. (Bhogal and Rudqvist, 2025, p.11). Taken together, these challenges show that organizational readiness is made up of multiple dimensions: data infrastructure, human capabilities, and constraints outside the organization's control. This makes a structured readiness assessment necessary, so that when an AI tool is selected, both its technical feasibility and its practical implementation within the current industrial environment are considered.

2.5 Synthesis from Literature Gaps to Framework Design

This chapter will discuss the findings of the literature review, along with their contribution to current research about cost engineering. This literature review presents a consistent basis for progress in thought. The previous sections have covered the following: cost engineering and its limitations, AI, and the organizational challenges to adopting AI. Cost engineering is a high-stakes activity that operates within the organization's procurement function. Traditional cost engineering methods have worked in gradually changing, stable, information-rich environments, but are not well-suited to the speed, volume, and complexity of modern automotive procurement. The current literature identifies two areas where more work is needed before AI can move from a concept to an established function in cost engineering: structured evaluation of AI tools and organizational readiness for adoption.

The first area where the literature shows a gap is in the evaluation of AI tools for use in a cost engineering and procurement program. There are five distinct AI use cases in cost engineering: cost estimating, supplier evaluation, contract analysis, spend analytics, and process automation. Therefore, organizations cannot simply choose to implement AI. Organizations must also evaluate their current technology capabilities to determine which AI tools can be implemented now and which will provide the most benefit moving forward. At present, there is no structured methodology for organizations to evaluate or implement AI tools within cost engineering or the procurement sector, as well as a lack of standardized design and development. Both issues create uncertainty for organizations regarding which AI tool they should select for implementation. An integrated TOE benchmark framework is proposed in Chapter 3 to assist organizations in evaluating the AI tools across six measurable factors within the three TOE dimensions.

The second key gap is contextual in that an organization's willingness and preparedness for AI adoption will vary from one organization to another and may not be based on their technological attributes alone. At Company X, Bhogal and Rudqvist (2025) characterize various strengths and challenges in the areas of data quality, technology infrastructure,

employee education, change management processes, leadership commitment, and collaboration with other organizations. Each of these characteristics can be a complementary support for the selection of AI tools and their feasibility for implementation. This gap is associated with the challenges faced by organizations, as discussed in Section 2.4, including working with limited data, dealing with employee resistance to change, and operating within imposed governance constraints. A framework that does not take these challenges into consideration will generate valid and feasible recommendations, but they would be impractical to execute. These challenges are addressed by integrating the readiness logic directly into the same six-factor benchmark as detailed in Chapter 3. The integrated approach treats the benchmark scores not only as a comparative ranking but also as a readiness signal across the three TOE dimensions, eliminating the need for a separate readiness assessment.

These two gaps are interdependent: benchmarking allows an organization to decide what to adopt, while readiness allows it to implement that choice in practice. Collectively, the two approaches provide a complete answer to the research problem. The integrated TOE benchmark addresses all three research questions through the same factor structure. RQ1 is addressed by mapping the AI tools and use cases relevant to cost engineering and procurement. RQ2 is addressed by comparing AI tool scores across Company X and its peer organizations on each factor using the integrated TOE benchmark. RQ3 is addressed by reading the benchmark scores as readiness signals and translating them into prioritized recommendations. This single-framework design ensures that the comparative ranking and readiness assessment use a consistent factor structure, rather than risking conceptual misalignment between two separate frameworks.

The framework presented in Chapter 3 is therefore not an external construct imported into the case, but the logical synthesis of the literature gaps identified in this chapter and the implications for making actionable AI adoption decisions in cost engineering.

3. Theoretical Framework

The objective of this study is to tackle the challenges outlined in Sections 2.3 and 2.4. Firstly, there is a lack of structured evaluation of AI tools within the cost engineering function across Technical Scope, Integration, Use Cases & Value, Workflow & Culture, External Drivers, and Ecosystem Readiness. Secondly, there is a lack of evaluation and recommendations for organizations to adopt these technologies. The two challenges are interlocked; the decision on which AI tool to implement is connected to the conditions that allow its implementation in the organization. This is why this study uses an integrated theoretical framework rather than two separate theoretical frameworks for benchmarking and tool assessment. The six factors considered in the integrated framework are the result of triangulating the findings from the literature review, the outcomes from the cost engineers and IT head interviews carried out at Company X, and the domain-specific selection made by Company X management. The factors are grouped by the structural logic of the TOE framework (Tornatzky, Fleischer and Chakrabarti, 1990; extended by Marhraoui, 2026). The TOE framework is used to structure the discussion and helps to identify the different contexts of adoption through the lens of Technological, Organizational, and Environmental perspectives. Two measurable factors are set within each of the TOE dimensions to accommodate the six integrated factors that are used to score, compare, and interpret AI tools in this study. This integrated design is achieved by the same factor structure, yielding the comparative tool ranking for RQ2 and the priority recommendation for RQ3.

3.1 The Integrated TOE Benchmark Framework

The first segment of the integrated framework aims to look at the benchmarking of AI tools that can be related to cost engineering use cases. Organizations face a great challenge in deciding among the vast range of AI tools and architectures on the market; as previously discussed in Section 2.5, a structured evaluation of available AI tools is required. This study was based on four existing frameworks: Artificial Intelligence Maturity Model (AIMM), Artificial Intelligence Capability Assessment Model (AI-CAM), Capability Maturity Model (CMM), and Cross-Industry Standard Process for Data Mining (CRISP-DM). These frameworks provide valuable lessons on the maturity and structure of AI deployments, but they focus solely on technology and do not discuss the challenge of choosing the right AI tool for a given organizational situation. They serve as a reference tool, but not as the main assessment criteria. In addition, none of these frameworks link tool-level scoring to organization-specific next actions, which is what Company X needs in practice.

The integrated TOE benchmark developed in this study is therefore designed to do both: rank the tools on a common factor set and translate the same factor-level evidence into a prioritized recommendation for Company X, which a single maturity framework or a standalone TOE assessment could not deliver. Use cases from functions around cost engineering are included in one of the factors in the integrated TOE benchmarking. As noted by Balkan and Akyuz (2025, p. 2), the applications of AI and ML span various cost engineering processes, such as supplier selection, contract management, and cost prediction.

This wide range requires the selection of use cases that are deliverable as a result to Company X. The integrated framework measures AI tools in six measurable factors in the framework of the three TOE dimensions to address the gap in existing maturity frameworks. The technical scope and Integration factors are grouped under the Technology dimension. Use cases & Value as well as Workflow & Culture factors are grouped under the Organization dimension. External Drivers and Ecosystem Readiness factors are under the Environment dimension. The construction process is described in Section 4.5, and each factor has four maturity levels (L1-L4) that are empirically anchored in evidence from interviews, literature reviews, and Company X management's selection priority. The six factors are not independent of each other; each reflects a different dimension of an AI tool's adoption context and its state of use within the organization. All the factors are validated together to attain the right decision of priority for adopting an AI tool based on practical evidence.

3.1.1 Technology Scope

The technical scope highlights the complexity of the tools, as they were designed to use AI. There are several AI techniques available in the market, each having a different type of problem and data structure, including ML, NN, NLP, Large Language Model (LLM), and GenAI. To ensure that an application is in line with the technology and business needs of the organization that is adopting the application, knowledge of the underlying technology is essential. This factor is reverse-scored for the technical factor only (L1 = 4, L2 = 3, L3 = 2, L4 = 1), since the empirical reasoning indicates that the less complex the architecture, the greater the value to Company X in its current readiness state, considering all other factors. Simpler tools are easier to adopt when integration capacity is limited, and no additional skills are required for the team, and users build trust in them faster because the lower complexity makes the cost logic easier to explain. The opposite reading is also possible: a higher level of AI architecture can help in more complex tasks, but for Company X in its current state, the relationship works in reverse. This stance is supported by the interview data, as five of the six cost engineers and the IT head cited technology complexity as an adoption challenge, instead of a measure of capability. This context-dependent reversal aligns with multi-criteria decision analysis principles, in which the direction of factors is adjusted to the decision context rather than treated as universally fixed (Aytekin, 2021). The reversal is thus methodologically grounded as well as empirically derived.

3.1.2 Integration

The level of integration is a measure of how deeply an AI tool is integrated into the existing data systems, processes, and decision-making within the company. At automotive OEMs, cost information is spread across engineering tools, ERP systems, procurement data, supplier portals, and even legacy spreadsheets. An AI tool's ability to link these solutions is both a prerequisite and a big asset for the company's efficient operation. According to Marhraoui (2026, p.925), data, model, system, infrastructure, server, and network are all parts of TOE, and they all play a role in integration depth. The levels and ranking for integration follow a standard format (L1 = 1, L2 = 2, L3 = 3, L4 = 4).

3.1.3 Use Cases & Value

Use Cases & Value captures the broad scope of the AI use, as well as the measurable value delivered to cost engineering and related procurement processes. There are several different AI tools for each of the functions: cost estimation, supplier evaluation, procurement analytics, process automation, contract review, spend analytics, and negotiation support (Balkan and Akyuz, 2025, p.2). It's essential to map a tool to its use case to understand the value it brings to the team or commodity. Use Cases & Value is scored using standard rankings based on empirical evidence from interviews, with measurable outcomes that reduce cycle time, eliminate manual effort, or otherwise generate specific value claims, which score higher than those viewed as general productivity layers. One major point to note in this thesis is that the cost estimation tools, or tools related to cost engineering, are ranked first in the group, followed by tools related to other use cases that support cost engineering. This again proves the need to focus on case-specific requirements for Company X.

3.1.4 Workflow & Culture

Workflow & Culture captures the human and process readiness of the organization to absorb an AI tool in the cost engineering function. As discussed in Section 2.4, the organizational dimension of AI adoption is decisive; even the most technically sound application might not deliver intended value if it is deployed into a workflow where the ownership is unclear, training is absent, or the team is resistant to change. This factor evaluates the level of cost engineering ownership for the tool, the change management activity surrounding it, the upskilling and training delivered, and the integration of the tool into daily working practice. Workflow & Culture reflects the mindset of the people and the company itself, and uses the standard scoring level, which is observable in the interview evidence from Company X.

3.1.5 External Drivers

External Drivers reflect the regulatory, competitive, and strategic forces influencing the urgency of adoption. In all the automotive OEMs, there is substantial pressure coming into the picture in various forms, and the strategy to deal with it is translated into the tools that differ. This factor thus indicates whether the deployment of a tool is explicitly linked to a named external motivation such as group-level policies, General Data Protection Regulation (GDPR) compliance, or competitive positioning. There can also be a possibility to introduce a tool as a reaction to external pressure without providing a named motivation. The logic of scoring is standard for the external drivers.

3.1.6 Ecosystem Readiness

Ecosystem Readiness collects data, infrastructure, and partnership context details at Company X, supplier, and group levels that influence the access to data that AI tools require to operate. For example, the ML-based cost estimation input is the supplier-provided data, and if this data is missing, not standardized, or commercially unavailable, then the limitation

cascades up through the ML tool into its predictive reliability. The transparency of supplier data is therefore a prerequisite for any AI-based cost estimation, as discussed in Section 2.4, and the data quality of the entire ecosystem is the biggest environmental hurdle to the adoption of AI. Ecosystem Readiness scoring logic is standard, as with other factors.

3.2 Theoretical Underpinning of the TOE Framework

The integrated TOE framework is based on the early framework suggested by Tornatzky, Fleischer and Chakrabarti, (1990) and further developed for the AI adoption context (Flechsig, Anslinger, and Lasch, 2022; Riti, Bacali, and Abrudan, 2025; Marhraoui, 2026). Three reasons were considered for choosing the TOE framework. First, it is a study at the organizational level, which is the most appropriate unit of analysis for a case study on the implementation of AI within a single automotive OEM. Secondly, its multi-dimensional structure (separation of technological, organizational, and environmental contexts) offers an adequate structural coherence to integrate the comparative benchmarking function within the readiness assessment for priority recommendation in one framework. Thirdly, it has shown its applicability for understanding the role of various factors in shaping the viability of new technology in an industrial setting; e.g., Riti, Bacali, and Abrudan, (2025) indicate that technological readiness and organizational support determine the extent and depth of new technologies being adopted in an industrial organization.

3.2.1 Technological Dimension

The technological dimension concerns the availability, capability, and compatibility of current technology for adopting AI techniques. This context is realized in the Technical Scope and Integration factors within the approach. If we consider the cost engineering and procurement at Company X, it involves assessing the data availability, depth of system integration, and analytical infrastructure capability. AI tools require good-quality data to be reliable, which is especially the case at the internal data level, and then examined with external supplier data. The quality of the available data determines how well AI tools can access and process it, as well as their integration with legacy systems. In the context of AI adoption, Marhraoui (2026, p. 925) highlights three key technological aspects, namely data quality, system architecture, and the capabilities of the infrastructure, which are in line with the technical scope and integration factors outlined in this chapter.

3.2.2 Organizational Dimension

The organizational dimension focuses on internal factors that impact the process of implementing AI: Leadership, skills, governance, culture, and capacity of the functional team in which the AI is being implemented. This is realized as the Use Cases & Value and Workflow & Culture factors in the integrated framework. In the context of industrial companies, Marhraoui (2026, p.921) states that corporate cultural barriers are one of the main obstacles to the use of AI tools. Existing research indicates that employees have the potential but need enhancements in change management and adaptability skills when it comes to

Company X to leverage AI effectively. This study builds on this finding by exploring the organizational state of the tool-level, through the two organizational factors of the integrated framework, to arrive at a more fine-grained reading than that of the functional level.

3.2.3 Environmental Dimension

The environmental dimension reflects external influences that impact the company's plan of using AI, such as regulations, industry pressure, supplier capabilities, and the surrounding network of customers, suppliers, and partners. In the integrated framework, the factors are realized in terms of External Drivers and the Ecosystem Readiness perspective. Cost engineering and procurement is a key area where supplier readiness is critical. In this context, compliance, transparency, and accountability are at the heart of the issue, and organizations must deal with a data-intensive environment in which commercial information is shared with external suppliers. The impact of other peer organizations and competitors in the digital transformation race is another external factor that affects the urgency of adoption. The two analytical factors, External Drivers and Ecosystem Readiness, are to be documented for understanding the strategic drivers that influence the adoption, and for the need for data infrastructure that hinders the adoption from an environmental view.

3.2.4 Previous Research Using the TOE Framework

Previous research has adopted the TOE framework to explore and understand the adoption of, and barriers to, digital technologies, including AI tools, in different industrial contexts. Flechsig, Anslinger, and Lasch (2022) suggest that the TOE framework would be a valid theoretical framework for studying contextual factors that affect the adoption of IT innovations. In the TOE, analysis can be focused on the firm level to describe the relationship between technological maturity, governance, and market dynamics, including the depth of AI adoption in the portfolio (Riti, Bacali, and Abrudan, 2025, p. 5). The TOE framework has also been extended to consider the organization and specific factors related to individual case studies of AI tools. The framework is demonstrated in a variety of settings ranging from e-commerce and retail, through ERP adoption to Small and Medium-sized Enterprises (SMEs), large companies, and even in the automotive OEM industry with AI adoption, which is clearly defined by Marhraoui (2026, p.922). The emphasis here is on the need for a developed TOE framework in the AI era, but across these applications, benchmarking and readiness assessment for priority recommendation have been treated as two separate instruments. This study brings together in a single factor structure.

3.2.5 Use of the Integrated TOE Framework for this Study

The study analyzes the adoption of AI tools and use cases in cost engineering and procurement for Company X. The TOE framework in this study structures both the comparative ranking and the readiness assessment priority lens, with six measurable factors integrated into the framework. The Technology dimension, with Technical Scope and

Integration, assesses how well each AI tool performs in terms of its architectural capacity and the extent of integration. The Organization dimension, having Use Cases & Value and Workflow & Culture, rates the value added by the deployed tools and the context in which they work. The Environment dimension, with External Drivers and Ecosystem Readiness, examines the strategic anchoring of the tool's execution, as well as the conditions of the supplier ecosystem around the tool. Combined, these factors provide a structured framework to assess the viability of AI tool implementation in Company X and the potential value of specific tools in the future.

This thesis contributes to the methodological economy that does not exist in the previous TOE framework, since it offers integration of the tool-level benchmark and recommendations for the AI tool assessment in a single framework. Each factor is scored based on the tool's contribution level from L1 to L4, considering the organization's priority, which reveals the order of adoption priority. This translates the analytical link from RQ2 to RQ3: the TOE readiness signals are read through the dimension structure of the benchmark score. A separate readiness assessment is not necessary, as the readiness picture is apparent from the scores on the benchmark. The recommendation matrix in Chapter 6 operationalizes this dual reading by translating the benchmark scores into prioritized actions for Company X. Table 1 below shows the six defined factors through the TOE lens, with each definition derived from the triangulation of the literature review, the cost engineers and IT head interviews at Company X, and the management requirements at Company X.

Level	Technology Dimension (T)		
	Maturity Stage	**Technical Scope**	**Integration**
L1	Ad-hoc	Basic Rule-based systems / Excel Macros.	Standalone; siloed data.
L2	Pilot	ML for specific price regressions.	Application Programming Interface (API) based connections but not "live" in ERP.
L3	Integrated	NLP for contract analysis; GenAI for supplier queries.	Deeply embedded in Product Lifecycle Management (PLM) /ERP/ Supplier Relationship Management (SRM) systems.
L4	Strategic	Multi-modal GenAI and Predictive Prescriptive Analytics.	Seamless data thread from R&D to Procurement.
Level	**Organization Dimension (O)**		
	Maturity Stage	**Use Cases & Value**	**Workflow & Culture**
L1	Exploratory	High-effort/Low-frequency use cases. The organization cannot quantify or consistently demonstrate the value of use cases.	Resistance to change; "Black box" scepticism.
L2	Defining	CBDs for simple parts. Measurable cycle time reduction.	Process changes initiated. Specific people to lead, advocate for, and drive AI adoption have been designated.
L3	Established	Predictive spend analytics; high accuracy in "Should-Costing."	Data-driven culture; AI-native workflows are standard.
L4	Leading	Real-time global cost optimization. AI drives negotiation strategy.	Centre of Excellence (CoE) scales lessons across organizations
Level	**Environment Dimension (E)**		
	Maturity Stage	**External Drivers**	**Ecosystem Readiness**
L1	Reactive	Adopting AI only because competitors are.	Low supplier data transparency; paper based.
L2	Compliant	Focused on regulatory reporting (e.g., Environmental, Social and Governance (ESG) /Supply Chain Act).	Digital portals exist; data quality is inconsistent.
L3	Proactive	Driving AI to combat volatility and inflation.	Standardized data exchange with Tier-1 suppliers.
L4	Orchestrating	Setting industry standards for AI in Procurement.	Full ecosystem visibility; collaborative AI-costing.

Table 1. Six factors in the TOE framework and their level definitions (Source: Authors)

4. Research Methodology

This chapter explains how the study was designed and conducted. It presents the research philosophy, the overall research design, the rationale for case selection, and the data collection and analysis methods used in the study. Research quality, ethical considerations, and limitations are addressed in the final sections, which ensure the credibility of the methods performed. The aim is to demonstrate that the methodological choices made are appropriate for the research questions and consistent with the theoretical framework presented in Chapter 3.

4.1 Research Philosophy

The way a researcher understands what reality is and how it can be known shapes the methodological section that follows. This study is concerned with how organizations perceive various AI tools and their adoption in cost engineering, a phenomenon that does not exist as an objective reality but is instead shaped differently by every organizational context it inhabits. The view of different AI tools and their adoption at Company X is influenced by its specific data infrastructure, team culture, leadership priorities, and process history. The same technology may be mature and impactful in one organization but entirely impractical in another. For this reason, the study takes a largely constructivist ontological position, treating the adoption of AI tools as context-dependent rather than as a single shared reality that can be measured in the same way across all organizations. In line with the pragmatic stance set out below, this does not rule out structured comparison; the benchmark scores used later are read as a practical, context-aware ordering of the tools rather than as objective measures of a fixed reality. The meaning is constructed through experiences and decisions by the people and organizations involved. (Creswell, 2009, p.8)

This ontological stance directly informs the epistemological approach required for a context-dependent study, in which reality is understood as socially constructed through human practice and knowledge is generated mainly through interpretation rather than measurement alone. Accordingly, the study adopts a pragmatic interpretive stance. It remains interpretive because it relies on how practitioners describe and make sense of AI tools within their own organizational settings. At the same time, it is pragmatic because the study is driven by a practical problem, namely which AI tools Company X should adopt, and it therefore uses structured benchmarking, scoring, and ranking as analytical instruments that turn these interpretations into actionable recommendations rather than as claims to objective measurement. In practice, this entails gaining close access to the organizational settings in which decisions regarding AI tools are made, engaging with practitioners' accounts of their experiences, and interpreting these accounts in relation to one another and to external evidence. As Czarniawska (2000, p. 23) argues, meaning in organizational research is not extracted from data as an inherent property, but is constructed through the act of interpretation itself. The researcher is therefore an active participant in this process rather than a neutral observer. This emphasis on interpretation as central to the research process distinguishes interpretivist inquiry from approaches that treat data as self-explanatory. Taken

together, these philosophical positions justify the qualitative research design adopted in this study. The aim is not to produce statistically generalizable findings on AI tools and adoption across all automotive OEMs, but rather to generate context-specific insights into organizational practices and challenges, thereby providing meaningful, in-depth understanding and informing potential solutions.

4.2 Research Design

Given the pragmatic interpretive stance described above, this study adopts a qualitative case study design with an embedded external benchmarking component. Company X serves as the primary case, and the study seeks to understand the AI tools and their adoption challenges in the cost engineering function as it is navigated within this organization. The case study is not treated here as a limitation within this scope, since it is a methodological choice that enables the deep, contextual knowledge this research requires. The case study is particularly well-suited to research situations under which the phenomenon being studied is inseparable from the context in which it occurs (Flyvbjerg, 2006, p. 223). This is the current situation at Company X, where there are a lot of AI tools in the market, and its adoption strategies cannot be understood by abstracting away the organizational conditions, structures, and group ecosystem. Flyvbjerg (2006, p. 224) points out that case studies do not produce inferior knowledge simply because they do not generalize statistically; they produce a different knowledge base grounded in particularity, context, and interpretive depth. This is the purpose of this study, aiming to generate knowledge for Company X.

The study is structured around a single integrated theoretical framework that addresses the three research questions. The framework combines the TOE adoption logic (Tornatzky, Fleischer and Chakrabarti, 1990; Marhraoui, 2026) with six measurable factors derived through triangulation across literature, cost engineers, IT head interviews of Company X, and management requirements of Company X. The same factor set produces the comparative tool ranking that answers RQ2 and the readiness signals that inform the prioritized recommendation for RQ3. Treating benchmarking and readiness as two separate frameworks would have introduced conceptual duplication and potential conflict between two scoring schemes; the integrated design ensures internal consistency and analytical economy. Both frameworks function not as descriptive checklists but as analytical instruments that structure how the empirical material is read, coded, and interpreted throughout the study. Several existing frameworks (AIMM, AI-CAM, CMM, CRISP-DM) were reviewed but found unsuitable for the specific purpose of this research.

This limitation motivated the development of an integrated TOE benchmark framework with six measurable factors, as presented in Chapter 3. The study builds on prior research conducted by Bhogal and Rudqvist (2025), which mapped the internal cost engineering practices at Company X and identified process gaps, along with a readiness assessment for digital transformation. These findings serve as an empirical foundation that complements the integrated TOE benchmark in this thesis, where Bhogal and Rudqvist mapped the functional-level cost engineering process gaps, and this thesis maps the tool-level AI adoption state.

Together, the two studies provide complementary readings of the AI adoption challenge at Company X, supporting the prioritization of actionable recommendations. Figure 1 provides a detailed explanation of the research design, participants, and their relation to each research question.

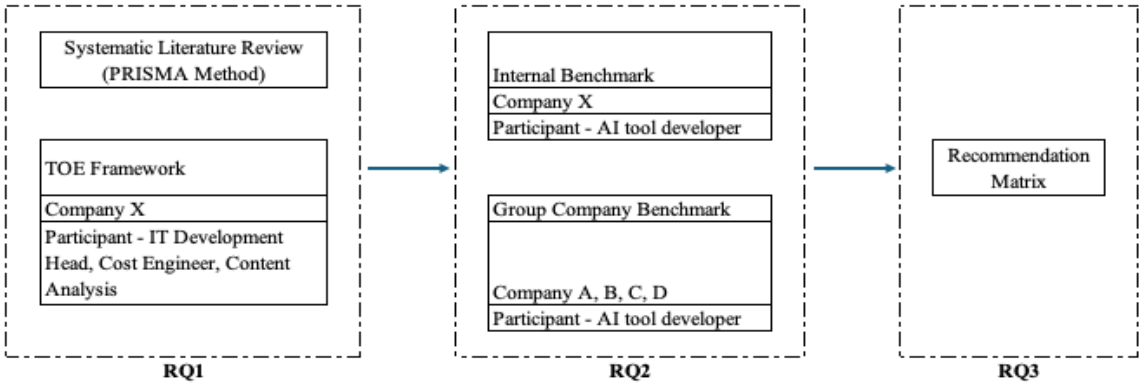


Figure 1. Detailed research design (Source: Authors)

4.3 Case Selection and Research Context

Company X is chosen as the primary case because it represents an automotive OEM at an active transition point from traditional, manually driven work methods towards a more data-supported and AI-enabled way of working. The research conducted by Bhogal and Rudqvist (2025) found that Company X is motivated and has some early foundational capabilities to support this type of transition, making this study a good example of combining benchmarking insights with additional organizational ability to form an empirical framework for AI adoption. Therefore, this is a single case study where the key to case selection is finding one that is relevant to the research question and has the potential to provide in-depth information about the issues being researched.

Furthermore, Company X has six internal AI tools at varying levels of maturity and use cases referred to throughout this thesis as Tools XA, XB, XC, XD, XE, and XF, and all of them are included in the benchmark. Tools XA and XB sit directly in the cost engineering workflow and address the should-costing or price estimation use cases, whereas other tools XC, XD, XE, and XF are related to other procurement use cases and functions (risk, payment terms, knowledge retrieval, and control tower analytics) but were included in the benchmark because they make part of the contribution in cost engineering work. Section 4.4.2 lists the corresponding tool and interview details.

The external benchmarking aspect of the study includes four group companies within the bigger umbrella, referred to as Company A, Company B, Company C, and Company D. The use of four group companies in this analysis was based on purposive sampling, which is a non-probability sampling method of selecting subjects based on the relevance of certain characteristics to the research question. In this instance, the group companies were selected

based on their demonstrated level of active engagement with AI implementation in cost engineering and procurement. The cost engineering department at Company X provided information on those group companies that have demonstrated active engagement with AI and, therefore, are viable candidates for use in this aspect of the investigation and are information-rich, which directly contributes to the benchmarking goals of this study. These organizations operate under a shared corporate-group structure that provides common governance, data standards, and platform alignment, which is the analytical foundation for the intra-group benchmarking and recommendation logic applied throughout the study.

The four companies were also validated for research purposes through snowball sampling, where people who work directly with one of these organizations were asked if they knew of any other peer organizations doing similar work in this area, resulting in no additional peer organizations related to this study. The rationale behind selecting benchmarking group companies versus competitors is based on the context of the study. All group companies share common data structures, technical platforms, and overall organizational standards. Thus, the AI tools and practices that one group company adopts will have greater potential for real-world transferability and applicability across each of the other member companies than do the same types of tools adopted by a competitor that is not part of the group. As such, for this study, intra-group transferability serves not only as a convenience but also as a critical aspect of practical concern in providing actionable recommendations for Company X. As a result of this finding, there is an increase in the external validity of the study's results, and its use can be applied to projects undertaken in a group context.

Furthermore, benchmarking against the competitors would raise ethical concerns regarding the use of commercially sensitive cost data, supplier pricing information, and negotiation strategies that organizations outside the group ecosystem do not have the consent to share for research purposes. Intra-group benchmarking avoids this ethical complication because the shared governance and data standards within the group provide a legitimate basis for cross-company comparison without the confidentiality issues that would arise with external competitors.

In addition to the peer organizations, the study uses a systematic literature review and publicly accessible industry materials to examine AI deployment strategies at other automotive OEMs for benchmarking purposes. Company X internal AI projects were identified by the IT head through interview and have been included as primary data on the state of AI maturity within Company X. These projects informed the TOE-based assessment of AI readiness by providing current examples of AI utilization in Company X. Combining these three data sources – the semi-structured interviews, the tool benchmarking evidence, and the systematic literature review enables a data triangulation that strengthens the reliability of the results and forms the empirical foundation for the analysis structure. Though insights from the other OEMs provide a comprehensive view of the industry, which was broad, they were not fully aligned with the scope of the Company X's cost engineering function. Their direct relevance seems to vary in organizational structure and tool usage. So, the primary weight is given to data shared among the chosen peer organizations, as they all

have comparable methods, systems, processes, and tools that are generally suited to Company X, ensuring the results are usable and align with the fact that the companies are doing the same types of work.

4.4 Data Collection

This section describes how the empirical material for this study was gathered through three complementary methods, namely a systematic literature review, semi-structured interviews, and a combination of Company X internal document analysis with an observation study. Each method serves as a basis for understanding the problem in depth and the reasoning for why this specific type of methodology is used for each study. This increases the credibility of the findings by mutually reinforcing perspectives for understanding AI tools and their adoption in the cost engineering function.

4.4.1 Systematic Literature Review

The method of finding relevant literature follows a systematic approach structured around three sequential phases: keyword analysis, keyword mapping, and the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) selection process. This PRISMA guideline will lead to more transparent, complete, and accurate reporting of systematic reviews, thus supporting evidence-based decision-making (Page et al., 2021).

The first step and challenge in reviewing the literature is finding the right words to describe the requirements for Company X, which, in this case, concern AI, cost engineering, and automotive companies. With this as a base, a separate set of terms is derived for all three search strings to avoid conceptual overlap. This keyword strategy is used to make sure all variations are captured, and the search includes all the important research articles systematically for the study. Each of the search strings is organized into Boolean operations, combining the first three strings that are mandatory terms for the search using AND operators, while the final optional search string relates to the OR operator. Each search string is arranged in the order of priority, and the keyword analysis is done with the help of knowledge from the authors, the cost engineering team of Company X, and generic terms used for the topic from internet searches. Refer to Table 2 for the summary of search strings gathered, their placement with respect to priority, and the Boolean operations.

The keyword analysis resulted in several terms for each search string, yielding a range of permutations and combinations. To standardize the process and to make the search practical, the Scopus database is selected, and all different combinations of search strings are executed. There are many other databases used in article searches, as detailed in Table 4, but given the large number of databases, Scopus is selected for preliminary analysis. A specific, limited keyword is fixed for each search string based on the highest number of relevant articles in the Scopus search results. Table 3 below shows the final keywords used for each search string, with every database used in this study for gathering research articles.

Search String 1	Artificial intelligence / AI integration / AI / Automation / AI tools / AI techniques / AI initiatives / AI adoption / New product development.
(AND)	
Search String 2	Cost impact / Calculation / Cost breakdown / Price reduction / Procurement / Cost Engineering / Cost / Cost drivers / Cost management / Optimized price.
(AND)	
Search String 3	Automotive / Heavy vehicle manufacturing / Buses / Auto makers / Manufacturing.
(OR)	
Search String 4	Use case / Risk analysis / Price prediction / Metrics / KPI / Sustainable / Strategy / Change management / Framework / Prerequisites / Recommended actionable steps / Case study / Analysis / Success patterns / Competitors / Maturity level / Performance indicators / Practical recommendations / Goal / Benchmark / Operational risk / Challenges / Negotiation.

Table 2. Keywords supporting the literature search (Source: Authors)

Search string 1	Artificial intelligence / AI / AI adoption
(AND)	
Search string 2	Cost / Cost engineering / Procurement
(AND)	
Search string 3	Automotive
(OR)	
Search string 4	Use cases / Outcome / Metrics / Risk analysis / Case study / Benchmarking

Table 3. Final search string keywords used for research articles (Source: Authors)

To cross-verify whether the finalized search string is connected to all relevant articles, the result JavaScript Object Notation (JSON) file from the Scopus search using the keywords from Table 3 is downloaded and analyzed using the VOS viewer software. This shows the connections between all keywords, as supported by prior studies using VOS viewer for bibliometric mapping (Mujahed, Feng and Wang, 2026). The exploded map in Figure 2 shows that the search terms, such as artificial intelligence, cost engineering, automotive, machine learning, supplier evaluation, should-costing, spend analytics, and manufacturing, are closely connected. This confirms that the search string covers both technical and cost engineering terms used in the literature study of AI in cost engineering.

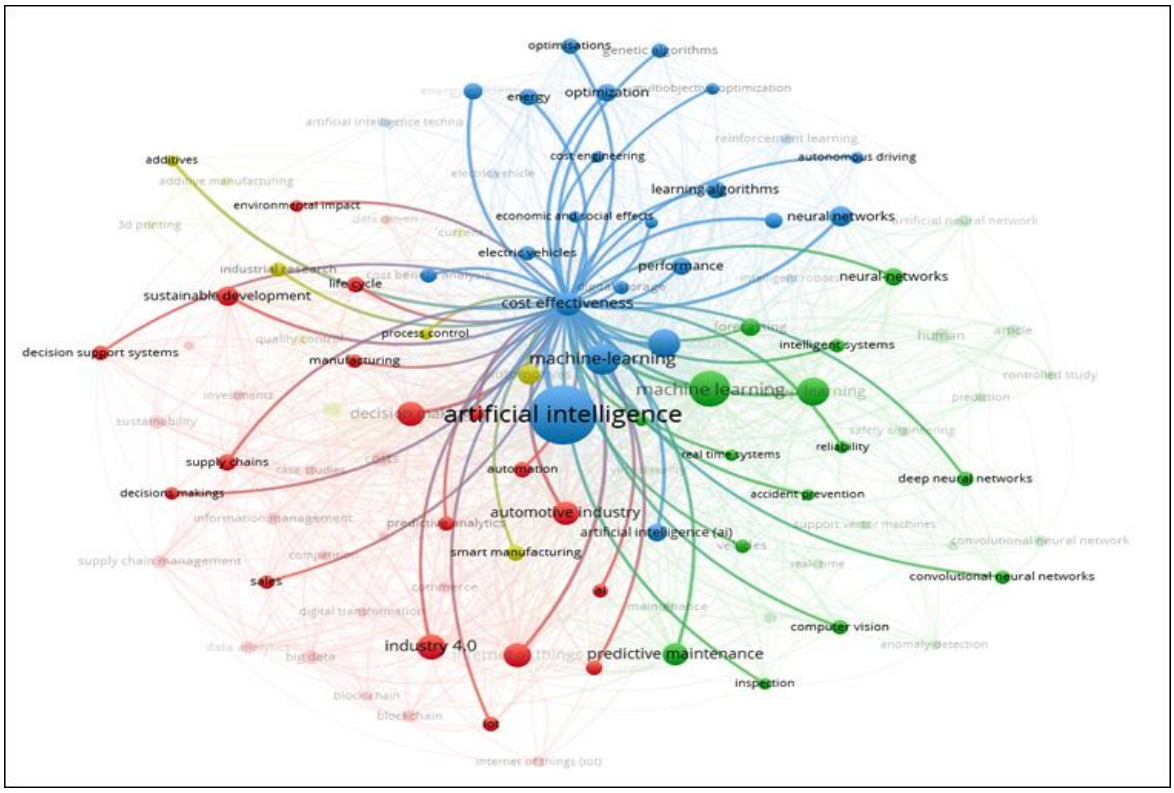


Figure 2. Exploded keyword map produced with VOS viewer software (Source: Authors)

The PRISMA framework provides a structured approach to the literature search and selection process. PRISMA was chosen since it is the recognized standard for article reviews in both engineering and management research, ensuring the selection process is reproducible, transparent, and bias-free. The PRISMA process unfolded into four stages: identification, screening, eligibility, and inclusion. The identification stage involved running the Boolean search strings (only the AND operation) listed in Table 3 across various databases: Scopus, Google Scholar, Uppsala University Library, KTH University Library, and ABI/INFORM. The database search was carried out in the same order, resulting in an initial pool of articles. Inclusion criteria were applied: Condition One (C1), English language from the year 2020 and peer-reviewed; and Condition Two (C2) was applied, which contains the final search string with an OR operation.

At stage two, there were 862 articles found with the search, and further, the titles were screened, and 104 relevant articles were downloaded by the authors. Stage three involves assessing the eligibility of these 104 studies by reviewing the abstract, conclusion, introduction, and key content, tables, and figures. The checklist used for selection involves AI relevance, cost engineering, and the automotive or manufacturing sector. Of the 104 articles entering eligibility, 55 were excluded during full-text screening for failing one or more checklist criteria, leaving 49 articles for inclusion review. The final stage involves the inclusion of the literature for the study and the elimination of any duplicate literature, since two different authors were involved in the selection process. This resulted in eliminating 7 duplicate studies and arriving at 42 articles, which were studied completely by keeping the

six benchmarking factors of the integrated TOE benchmark framework: Technical Scope, Integration, Use cases & Value, Workflow & Culture, External Drivers, and Ecosystem Readiness as the analytical priority during paper review. Table 4 presents detailed information on the PRISMA approach, stages, activities performed, database searches, and the number of studies.

Stage	Activity Performed	Database Name				Total Literature (Numbers)
		Scopus	Google Scholar	Uppsala Library and Diva Portal	KTH Library and ABI/INFORM	
Identification	Keyword search string	1492	9530	1541	1853	14416
Screening	C1 - English language, from year 2020, Peer reviewed	1395	525	399	1230	3549
	C2 - Additional keyword	218	180	254	210	862
Eligibility	Manual abstract and introduction study	19	27	35	23	104
Included	Full article study	-				**42**

Table 4. Literature selection process using the PRISMA method (Source: Authors)

4.4.2 Semi-Structured Interviews

The primary empirical data are collected using semi-structured interviews from Company X and other peer organizations (A, B, C, D) to understand the various AI tools used for cost estimation and other use cases related to cost engineering. As Creswell (2009, p. 178) points out, qualitative research involves purposefully selecting participants who will best help the researcher understand the problem and research question. For this reason, purposive sampling is used, i.e., the cost engineers at Company X are approached to reach out to the relevant people at Companies A, B, C, and D who are working on AI tool development with a cost-estimation scope. Further, snowball sampling is used on each interviewee to check the relevant people who can provide new and additional insights. A detailed summary of the interviewee list and sampling method is shown in Table 5 below. The aim of a semi-structured approach is to capture interviewees' experiences and make use of the information from a broad perspective, rather than based on predetermined hypotheses. The interview question was designed to be flexible enough to allow the interviewees to introduce perspectives and dimensions that the researchers may not have anticipated, while still maintaining enough structure to ensure comparability across interviewees and alignment with

the analytical frameworks. This is also supported by Creswell (2009) statement that, in qualitative inquiry, the researcher is the key instrument for data collection, and the design of the interview should support an active interpretive role rather than constrain it to answer categories.

A total of 19 semi-structured interviews were conducted between February and April 2026, of which 17 interviews are included in the final qualitative analysis. The remaining two were exploratory snowball interviews used to identify additional peer-company contacts and did not produce any tool or factor-specific data relevant to the benchmark scope. The 17 analyzed interviews are distributed as follows: Seven Company X internal interviews (six cost engineers and one IT head), six Company X tool interviews (one for each Tools XA-XF), and four peer organization tool interviews (one from each Companies A, B, C, and D). The five different companies in the interview count (Company X, A, B, C, D) for the AI tool are justified by the availability of AI tool users from a cost estimation perspective, meaning that out of all other peer organizations, there are only four known companies working with AI development in cost estimation, and the fifth being Company X itself. This data forms the basis for the integrated TOE benchmarking framework.

The aim of this part of the interview with the cost engineering team is to understand the organization-wide perspective on AI use in the cost estimation process, which contributes to the six factors of the TOE dimension and its definition for ranking. The interviewees are seven in total: one from the IT head, five cost engineers, and one senior cost engineer, all from Company X. The IT head interview serves two purposes: it feeds the benchmark coding (where the IT head's evidence informs Technology and Environment factor levels for Company X tools) and the Technological dimension of the TOE dimension. This dual coding is appropriate because the IT head holds the only enterprise-level visibility on platform capability, integration roadmap, and group-level data governance information that the cost engineers cannot fully access from their operational position, and that no single tool interview captures. The interview revealed four active internal AI projects, including a cost estimation initiative that contributes to both benchmark scoring and the TOE dimension analysis. Each of the interviews lasted between 45 and 60 minutes, and the session was conducted both online via Microsoft Teams software and in person, depending on the participant's convenience and availability. All interviews were audio recorded in Teams with the informed consent of interviewees, and the transcription was carried out using the automated transcription feature within Microsoft Teams, after which the transcripts were reviewed and corrected manually if needed to ensure accuracy.

Company	Tool	Department	Interviews		Duration (minutes)
			Numbers	Mode	
A	A	User	1	Online	45 - 60
B	B	Tool developer	1	Online	
C	C	Tool developer	1	Online	
D	D	Tool developer	1	Online	
X	XA	User	1	Online	
	XB	Tool developer	1	Online	
	XC	User	1	Online	
	XD	User	1	Online	
	XE	User	1	In person	
	XF	User	1	Online	
	-	IT head	1	In person	
	-	Cost Engineer	5	In person	
	-	Senior Cost Engineer	1	In person	
Total			**17**		

Table 5. Interviewee's list and details on interview time and mode (Source: Authors)

4.4.3 Observational and Document Study

To understand how the cost engineering team works in day-to-day operations, the observational study was conducted to experience it directly rather than to hear about it from others. From January through April 2026, a total of thirteen weekly group meetings were attended. This approach outlined how people are assigned work, how progress/updates are tracked with the team, how disagreements or problems are handled, the types of tools used within the team, and the team's improvement points. The meeting was attended in person, and field notes were taken during each meeting, including minutes. The structure of notes followed: what happened, what people said, team collaboration, who communicated with whom, what decisions were made, and what frustrations or improvement ideas emerged.

Alongside the meetings, the internal documents that the cost engineering department holds are reviewed. The primary content of the document involved the existing workflow diagram that laid out the stages of the cost engineering process, written procedures describing how things were supposed to be executed, and Company X's future in terms of technology and automation. Seminars and workshops were also held via online meetings to provide education on AI technology, its uses, and how it works in real time. According to Creswell (2009), document analysis is valuable because it enables researchers to obtain the language and words of participants and to access information at a time that is convenient, making it a useful source of information. This gave clarity on the difference between what Company X plans and how things are happening. All these strategic documents made the requirements

clear from a leadership perspective for the future. Since it is a standard document, it is very important for training new employees and for people who need an understanding of how basic things work in cost engineering and at the organizational level.

Due to the sensitive and confidential nature of the documents at Company X, these materials are not reproduced in the appendices. The observational and document insights are part of empirical analysis and understanding for authors about the cost engineering workflow, which was experienced throughout the engagement over the study period.

4.5 Data Analysis

The empirical material is gathered through the thematic analysis of the interviews with Company X, A, B, C, and D. As described in 4.2, the research design uses a single integrated TOE benchmark framework. The three interview guides: Appendix A for cost engineers, Appendix B for the IT head, and Appendix C for AI tool developers/users, were formulated to capture the research questions' requirements. The interview data were analyzed using the six-phase thematic analysis process described by Bell, Bryman, and Harley (2022), based on the original framework of Braun and Clarke (2006): familiarization, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and writing up the findings. The initial stage of familiarization is important to prevent any premature application of categories that may not fit the data required for analysis. Following the familiarization, the interview material was coded deductively using the six measurable factors of the integrated TOE benchmark framework: Technical Scope, Integration, Use Cases & Value, Workflow & Culture, External Drivers, and Ecosystem Readiness. In addition, the seven Company X internal interviews were coded inductively, allowing additional themes relevant to organizational AI readiness to emerge from the data even where they did not fit the benchmark factor structure.

Within the broader six-phase thematic analysis, the level assignment for each benchmarked tool follows a five-step sub-procedure that operates within Phase 2 (generating initial codes). This sub-procedure converts coded interview evidence into the numerical scores used in the benchmark ranking in Table 7. Level assignment for each tool on each factor follows a five-step pipeline that ensures transparency and replicability. Step one, thematic analysis, identifies the exact words and phrases interviewees used to describe the tool's state on each factor. Step two: keywords are extracted from these passages, for example, 'fuzzy matching', 'standalone', 'pilot', or 'a limited test group'. Step three: keywords are matched to the Key Table, which maps vocabulary to the relevant factor. Step four: the matched factor is compared against the Level Definition Table, where L1–L4 criteria are anchored in actual interviewee quotations from the Company X internal interviews, and the level that best matches is assigned. Step five: the level is converted into a numerical weight for cross-tool comparison. Five of the six factors use forward scoring (L1=1, L2=2, L3=3, L4=4) where higher levels indicate higher value. The Technical Scope factor alone uses reversed scoring (L1=4, L2=3, L3=2, L4=1) because the cost engineer and IT head interviews produced

converging empirical evidence that lower architectural complexity is more valuable for Company X at its current readiness state. Simpler tools are easier to adopt with limited integration capacity, less specialist skill on the team, and more immediate user trust. Five of six cost engineers and the IT head independently framed architectural complexity as a barrier rather than as a sign of capability, which is the convergent evidence base for the reversal. The conversion from qualitative levels to numerical weights is a structured ordinal device, not a measurement. Its purpose is twofold: first, to give Company X a clear order in which to engage with the ten tools across its own portfolio and the peer organizations, rather than facing them as an undifferentiated list; and second, to make a multi-factor reading of each tool visible in a way that a purely qualitative narrative would not identify. The interpretive substance of each level sits in the level definitions of Table 1 and in the factor justifications of Sections 5.3 to 5.5; the integer weights serve a comparison-navigation role for the cross-tool reading, not a performance-measurement role.

Factor selection was validated through triangulation across three sources that each play a structurally different role in the framework: literature as external grounding from published evidence, cost engineer and IT head interviews as the practitioner perspective from within the case, and management requirements as the organizational priority lens, and all three converge on the same six factors. The systematic literature review (Section 4.4.1) produced the theoretical constructs and prior empirical evidence for factor selection, drawing particularly on Bodendorf, Merkl and Franke, (2021), Hammann (2024), Balkan and Akyuz (2025), and Marhraoui (2026). The cost engineer interviews (six interviewees at Company X) revealed the operational requirements and pain points that ground each factor in practice: data quality, supplier transparency, integration depth, AI skepticism, regulatory pressure, and competitive urgency were each raised independently by multiple cost engineers. The management requirements (articulated by the IT head and the cost engineering team lead) established what Company X leadership considers strategically important for AI adoption decisions, including data governance, group-level coordination, and supplier-side integration. The convergence across these three structurally different perspectives, rather than across three independent measurements of the same construct, is the evidence base for factor validity. No factor was selected based on a single source alone, and each factor carries the weight of all three perspectives. This triangulation strategy follows established qualitative research principles described by Bell, Bryman and Harley (2022).

Each AI tool identified during the interview was assigned to these six factors, enabling a structured comparison of how different organizations approach the same technological opportunities. This deductive coding process made it possible to examine different patterns of agreement rather than treating each interview as a standalone account. Alongside this deductive coding, the analysis involved open themes that emerged inductively from the data. Several important insights did not fit the benchmarking requirement but addressed the TOE analysis. This integrated approach helped to capture the most out of the interviews for the framework. In the third and fourth phases, the initial codes were grouped to search for broader themes, and these themes were then reviewed by comparing them with the coded data to ensure they reflected the interview content. The fifth phase involves defining and

naming the themes to clarify what each theme captured and how it is related to the research questions, and the sixth phase consisted of writing up the findings in a structured form that links the analysis back to the theoretical frameworks.

The same six-phase thematic process is applied to the second stage of analysis, focused on the TOE framework from Company X interviews, thirteen team meetings, and internal process documents. The functional level readiness data produced by Bhogal and Rudqvist (2025) serves as a complementary functional level reference layer that allows tool-level benchmark scores to be read against their findings. The final stage of the analysis is carried out using the integrated TOE benchmark framework to obtain results indicating the practical and realistic conditions for Company X in adopting an AI tool for cost engineering. This synthesis creates a multi-stage recommendation matrix, examining each benchmarked AI tool against TOE insights and possible dimensions for development under different circumstances. Throughout the analysis, the two authors reviewed and cross-checked each other's interpretations at each phase. This peer examination reduced the influence of individual bias and ensured the analysis remained grounded in the empirical findings. Whenever the interpretations differed, the original source was revisited until a common understanding was reached.

4.6 Research Quality

The research quality was evaluated according to the criteria of credibility, transferability, dependability, and confirmability that are typically used in interpretive qualitative research (Bell, Bryman and Harley, 2022), which have a similar focus to validity and reliability but take into consideration the interpretative nature of the study. Credibility relates to whether the results represent the studied context accurately. Methodological triangulation, in the form of a systematic review of the literature, semi-structured interviews, and document analysis, establishes credibility by adding confidence in the findings through cross-validation. The use of structured frameworks for data interpretation also increases the credibility of the study, as they provide theoretical categories for interpreting the data. Finally, respondent validation was performed where possible to verify the accuracy of the interpretation. Transferability relates to the ability of other situations to use or apply the results generated by this study. The research does not attempt to provide a statistical generalization of the results but instead provides a framework for analytical transferability. It provides a structured process that can be customized by organizations that have similar benchmarking and TOE experiences. According to Flyvbjerg (2006, pp. 226-228), case studies can provide knowledge that is practically useful since they are situated in real life.

Dependability relates to the consistency of the overall research process. The dimensions of benchmarking and TOE are clearly stated, allowing for replication, while the structured interview guide facilitated consistency across all data collection. Documentation of the process and analysis provides a transparent audit trail. Confirmability concerns whether the researcher's bias has influenced the results. Confirmability is supported by supplying data from previous empirical research studies, confirming that the case study sample is

representative, and by continuously checking and comparing findings. This has minimized possible bias and has supported an objective viewpoint.

4.7 Ethical Considerations

To address the ethical obligations of protecting confidential business data and the rights of individual participants, four main ethical principles were applied to this study: not causing harm, obtaining informed consent, protecting individual participant privacy, and avoiding deception (Bell, Bryman and Harley, 2022, pp. 112-125). Company X is not identified or mentioned by name anywhere in this thesis, nor are the four peer organizations that have been benchmarked in the study referred to as Company A, Company B, Company C, and Company D. There is no disclosure of information that would allow any external reader to have insight or identify any of these entities. This anonymization also applies to organizational structures, tools, and procedures, which are provided at a level of generality that preserves sufficient information while safeguarding commercial sensitivity. Prior to their sessions, all interviewees gave their informed consent and were informed about the purpose and goal of the study, the voluntary nature of their involvement, the anonymization of their reporting contributions, and the use of audio recording solely for research. Additionally, participants were made aware of their freedom to withdraw from the study at any time without facing any consequences.

The decision to benchmark against group companies rather than external competitors also carries an ethical dimension. Accessing the technology, tools, commercially sensitive cost data, supplier pricing, and negotiation strategies from organizations outside the group ecosystem would raise confidentiality concerns that intra-group benchmarking avoids, given the shared governance standards within the group. This design choice, therefore, reflects an ethical constraint as well as a methodological selection.

All participants were aware that the observational study was being carried out, and the researchers remained impartial throughout, neither steering conversations nor sharing private observations outside of the study setting. All information gathered, including notes, interview recordings, and transcripts, was securely maintained, and only the researchers could access it for academic purposes. No conflicts of interest need to be disclosed. This investigation was carried out independently as part of a Uppsala University master's thesis.

4.8 Research Limitations

The methodological choices made in this study carry some limitations that are worth addressing. Purposive and snowball sampling helped to reach the right participants, but it also means that selection was based on accessibility and referrals rather than on a systematic mapping of expertise levels. Given the rapid development in the AI field, there may be practitioners or specialists with deeper knowledge who were not reached for the interview. This means that the broadest possible knowledge base may not have been fully captured.

Additionally, the benchmarking framework classification of AI tools depends on how interviewees describe them rather than on verified technical evaluations. The assessments of maturity level and performance impact carry subjective judgments, which creates bias in interpretation of the results. A further methodological note concerns triangulation: of the three sources that grounded factor selection, only the systematic literature review is fully external to Company X, whereas the cost engineer interviews and the management requirements both originate within the case organization. The design therefore provides convergence across three perspectives with different structures rather than across three fully independent sources, and the validity claim should be read accordingly.

The integration of the benchmarking data with the TOE framework for producing recommendations for Company X introduces a further layer of analytical assumption. The TOE assessment is based on information gathered from employees, official documents, and observations; however, Company X may have additional constraints that are not captured in this study. The recommendation matrix that combines two frameworks produces many possible permutations of the final output, and this creates uncertainty in how Company X or the authors will weight options at implementation. Even though the frameworks were developed and applied by the same two authors, which reduces individual bias, this does not eliminate the interpretive influence that comes from the design of the integrated TOE benchmark framework.

5. Empirical Findings

5.1 Introduction to Findings

This chapter presents the empirical findings derived from the integrated TOE benchmark assessment of ten AI tools across Company X and the four group peer companies, together with the qualitative findings derived from the cost engineer and IT head interviews at Company X. The structure of the chapter follows the three TOE dimensions framed in Chapter 3. Section 5.2 sets out the three-source triangulation logic that produced the specific level definitions in Table 1 of Chapter 3, since the connection between empirical evidence and those definitions is not captured in Chapters 3 and 4. The validity of the six benchmark factors rests on the convergence of three structurally different perspectives: the literature as external grounding, the cost engineer and IT head interviews as the practitioner perspective within the case, and the management requirements as the organizational priority lens. Sections 5.3, 5.4, and 5.5 present the findings for the Technology, Organization, and Environment dimensions, where each factor is read across the ten tools. Section 5.6 captures the synthesis of cross-dimension patterns that emerge from the factor-level analysis, and Section 5.7 consolidates the benchmark ranking that feeds the prioritized recommendation logic for Company X in Chapter 6.

Two methodological notes are necessary before exploring the presented findings. First, the four maturity levels (L1-L4) defined in Chapter 3 are given practitioner-friendly names: Ad-hoc, Pilot, Established, and Leading. These four terms are applied throughout this chapter when describing the level assigned to each tool on each factor. The level names appear inline where they are readable and as L1-L4 references when the focus is on the comparative structure. The level definitions and the five-step assignment pipeline are not repeated here; they remain part of Chapter 3 (Table 1) and Section 4.5. Second, the ten benchmarked tools split naturally into two analytical tiers consistent with the case selection criteria in Chapter 4. The first-tier tools (Tool A, Tool B, Tool C, Tool D, Tool XA, Tool XB) go directly into the cost estimation workflow. The second-tier tools (Tool XC, Tool XD, Tool XE, Tool XF) address procurement supporting functions that contribute to cost engineering indirectly. This tier distinction is introduced by the case selection scope and the requirement from Company X and is used in this chapter's cross-tool patterns and to structure the priority recommendation in Chapter 6.

In the empirical material that follows, the cost engineers at Company X are referred to as CE1 to CE6, and the IT head as ITH. The interviewees representing each AI tool are referred to as Interviewee A, Interviewee B, Interviewee C, and Interviewee D for the four peer organizations A, B, C, and D, respectively, and as Interviewee XA to Interviewee XF for the six Company X tools. The non-AI rule-based calculation environment used by the cost engineering team at Company X is referred to as the cost engineering platform throughout to distinguish it clearly from the AI tools under assessment.

5.2 Triangulation Logic and Factor Selection

Chapters 3 and 4 established that the six benchmark factors and their level definitions were derived through triangulation across three independent sources. What those chapters do not set out in detail is what each source contributed to the specific wording of each level cell in Table 1. That connection between the empirical evidence and the framework definitions is what this section provides, so that the reader can trace the origin of the level descriptions before encountering them applied to specific tools in Sections 5.3 through 5.5.

The first source, the systematic literature review, established the upper end of the level scale across all six factors. It produced the L2 through L4 cell descriptions by mapping the progression of architectural sophistication, integration depth, use case maturity, organizational absorption conditions, external strategic forces, and ecosystem governance that the published evidence associates with increasingly advanced AI deployment in cost engineering and procurement. For Technical Scope, the literature traced the architectural ladder from supervised ML and regression models (Bodendorf, Merkl and Franke, 2021; Hammann, 2024; Mirasçı and Aksoy, 2025) through NLP and deep learning combined with activity-based costing (Bodendorf and Franke, 2024; Spreitzenbarth, Bode and Stuckenschmidt, 2024) to multi-modal generative AI and predictive-prescriptive analytics at the leading edge, producing the L2, L3, and L4 cell descriptions respectively. For Integration, ERP and PLM connectivity as the value-realizing precondition (Spreitzenbarth, Bode and Stuckenschmidt, 2024; Althabatah et al., 2023) anchored L3, and a seamless data thread from R&D to procurement anchored L4.

For Use Cases and Value, should-costing and measurable cycle-time reduction anchored L2, predictive spend analytics and high-accuracy should-costing (Balkan and Akyuz, 2025) anchored L3, and real-time global cost optimization with an AI-driven negotiation strategy anchored L4. For Workflow and Culture, change management ownership, technology acceptance (Riti, Bacali and Abrudan, 2025), and managerial drivers in automotive manufacturing established the L3 data-driven culture and L4 CoE conditions. For External Drivers, competitive pressure and reactive imitation (Marhraoui, 2026) produced L1; regulatory compliance produced L2, proactive strategic deployment produced L3; and industry standard-setting produced L4. For Ecosystem Readiness, supplier data quality is the primary AI dependency (Bodendorf, Merkl, and Franke, 2021; Hammann, 2024; Wang, 2025), and unified data exchange and infrastructure governance are advanced forms, established at L3 and L4. What the literature alone could not produce was the L1 cells across all six factors and the calibration of where current practice at Company X sits within the scale. That required the second source.

The second source, the cost engineer and IT head interviews at Company X, contributed three things to the definition table. First, the L1 cell descriptions for each factor, grounded in the baseline conditions cost engineers experience daily: the standalone, manually operated calculation environment at Company X grounded the L1 Integration cell; the prevalence of rule-based working with no AI reasoning grounded the L1 Technical Scope cell; resistance

rooted in long tenure with existing tooling and skepticism toward opaque AI outputs grounded the L1 Workflow and Culture cell; reactive competitive pressure without a named strategic program grounded the L1 External Drivers cell; supplier data minimization and internal data inconsistency between business flows grounded the L1 Ecosystem Readiness cell; and the absence of quantifiable demonstrated value grounded the L1 Use Cases and Value cell. Second, the interviews calibrated the L2 descriptions by providing concrete operational thresholds. Measurable cycle-time reduction in RFQ processing was named as the threshold at which the AI use case value becomes demonstrable. Pilot-stage deployment with designated change advocates was named as the threshold at which workflow absorption begins. Third, the interviews provided the convergent empirical justification for the reverse-scoring decision on Technical Scope, the one factor where the scoring direction departs from the standard forward scale. Five of the six cost engineers and the IT head independently framed architectural complexity as an adoption barrier rather than as a capability indicator, establishing that simpler architectures carry higher immediate value to Company X at its current readiness state. Two themes emerged inductively from the CE interviews that did not map to any level cell but were too consistent across accounts to dismiss: Trust Calibration, the norm of treating AI output as a verified starting point rather than a final answer, and the Skill and Generational Divide, the uneven distribution of AI prompting literacy across the cost engineering team. Neither was confirmed by management as a standalone factor, but both are reported as named inductive themes in Section 5.4.2 and 5.5.1, respectively, because they condition how the level definitions translate into actual adoption behavior.

The third source, Company X management requirements articulated by the cost engineering team lead, confirmed which constructs surfaced by the first two sources are strategically relevant and which level descriptions reflect the organization’s current and aspirational position. Management validated all six factors, confirmed the reverse-scoring logic on Technical Scope, and identified Ecosystem Readiness as the single most constraining factor across Company X's current AI tool portfolio. They also acted as a filter, excluding Explainability, Data Architecture, ESG and Regulatory Compliance, Trust Calibration, and the Skill and Generational Divide. Each of these met one or two triangulation conditions but not all three, and was either absorbed into the nearest matching factor at the operational level or documented in Appendix D.

Table 6 below maps this process visually across the three TOE dimensions. Three observations from the table inform the analysis that follows. First, the six selected factors do not all carry equal literature support: Technical Scope, Use Cases & Value, and Workflow & Culture are anchored across all three sources, while Integration, External Drivers, and Ecosystem Readiness draw their primary anchoring from the cost engineer, IT head, and management perspectives, with the literature contributing only partial higher-level descriptions. This asymmetry is what justifies the integrated triangulation design, since the published literature alone does not capture the practitioner and organizational reality at the L1 baseline where Company X currently sits. Second, the additional factors raised by the literature – Algorithmic Depth and Explainability under Technology, and Technology Acceptance and Process Automation under Organization – are not absent from the

benchmark but are absorbed into the selected factors at the level of their operational expression: Explainability surfaces as black-box skepticism within Workflow and Culture (Section 5.4.2), and Technology Acceptance is absorbed into Workflow & Culture, while additional literature themes that do not map to the TOE dimensions are documented as raw coding in Appendix D. Third, Trust Calibration and the Skill and Generational Divide emerged inductively rather than deductively and are reported as inductive themes that nuance the deductive coding without disrupting the six-factor structure validated across all three sources.

TOE Dimension	Factors from Literature	Factors from CE1 to CE6 and ITH	Management Selection
Technology	Technical Scope	Technical Scope	Technical Scope
	-	Integration	Integration
	Algorithmic Depth	-	-
	Explainability	-	-
Organization	Use Cases & Value	Use Cases & Value	Use Cases & Value
	Technology Acceptance	Workflow & Culture	Workflow & Culture
	Process Automation	Negotiation Readiness	-
	-	Trust Calibration	-
Environment	-	External Drivers	External Drivers
	-	Ecosystem Readiness	Ecosystem Readiness
	-	Regulatory Constraints	-
	-	Skill and User Perspective	-

Table 6. TOE Factors from Literature review, thematic coding of cost engineers, IT head, and management selection from Company X (Source: Authors)

5.3 Technology Dimension Findings

The technology dimension pairs Technical Scope and Integration. As described in Chapter 3, Technical Scope uses reverse scoring for the benchmark scoring, with lower architectural complexity scoring higher, since the convergent evidence from five of six cost engineers and the IT head framed complexity as an adoption barrier rather than an indicator of capability. The two factors are paired analytically because they often move in opposite directions. More sophisticated architectures tend to encounter more integration friction in the fragmented cost data environment.

5.3.1 Technical Scope: Simplicity as an Adoption Advantage

Across the ten benchmarked tools, the architectural range extends from rule-based parametric logic to fine-tuned LLMs. The most architecturally advanced tools are Tool XA and Tool C, both LLM-based, and Tool XF, which is designed around generative AI but remains in a

conceptual stage. Tool B currently runs a fuzzy-matching algorithm, with LLM integration planned. At the simpler end, Tool XB is rule-based, and Tool A is a generative AI layer with no native connection to the specific cost calculation platforms. On the reverse scoring logic, the highest scores are obtained by the simplest tools. Tool B and Tool XB sit at L1 (Ad-hoc). Tool A, Tool D, Tool XC, Tool XD, and Tool XE sit at L2 (Pilot). Tool XA, Tool XF, and Tool C sit at L3 (Established). No tool reaches L4 in the dataset. Interviewee XB described the logic of intentional simplicity directly with the following statement.

"I use Tool XB, and that is not an AI at all. It's very simple mathematical logic. It's some kind of advanced Excel, basically, that I can do this coding in the background to build the models." – Interviewee XB

The same logic was supported from a different angle by the cost engineers, where the binding constraint on value is not capability but defending the calculation logic in supplier negotiations. CE4 captured this directly in the statement below:

"If a tool gives me a number and I cannot explain how it got there, I cannot take that into a negotiation. The supplier will question every line. I need to be able to stand behind the calculation". – CE4

CE3 reinforced the same point in terms of black-box skepticism. The data architecture and infrastructure capabilities are core technological aspects of AI adoption, and this extends that observation by showing, transparently, what is considered architectural value. The reverse scoring logic on technical scope encodes these findings into the framework. An observation emerged from the cost engineers' interviews that can be applied to formal scoring. CE2 reported individual use of GenAI, such as ChatGPT and Claude, for quick extraction of general information, which accounts for ninety to ninety-five percent of searches online for details, and this raises the expected value well above what mid-level tool deployment delivers.

"I am using ChatGPT and Claude personally for maybe ninety-five percent of my general data collection. It is an assistant, not a replacement, but the efficiency difference is enormous compared to doing it manually." – CE2.

This single-engineer L3 equivalent practice is not reflected in any tool-level score because it operates outside any sanctioned tool deployment. It is reported here because it indicates that the gap between what the organization rolls out and what individual engineers use is wider than a tool-only reading captures.

5.3.2 Integration: The Fragmented Systems

Integration findings reveal a structural constraint that is needed for the adoption of AI tools at Company X, more consequential than architectural complications. With one exception (Tool XD at L3), all benchmarked tools operate in conditions of partial or siloed integration. Tool

A's AI assistant has no integrated connection with the data at Company A. Tool D operates as a standalone tool with manual image upload. Tool XB's limited licenses cover only the designer population. Its lead describes Tool XA, the most architecturally advanced cost-estimation tool at Company X, as an "in-between" state:

"It's only searching for specific rows and specific data, extracting that, modifying the data, and presenting it in a better way. It doesn't feed anything back. It's not standalone, it's not integrated. So, it's in between." – Interviewee XA

The pattern reflects a structural feature of the cost engineering data environment that the Company X cost engineers describe in operational terms. CE5 described the calculation environment directly:

"The cost estimation platform is completely standalone. Every time I need data from another system, I export it, clean it, and import it manually. There is no live connection to anything. The AI sits on one side, my costing tool on the other, and I am the bridge." – CE5

CE1 added the same observation from the perspective of the AI assistant deployment at Company X, where the same structural disconnect applies:

"The AI assistant is on one collaboration platform, cost calculation software is somewhere else entirely. They do not talk to each other. So, I use one to think and the other to calculate, and I do the bridge myself in between." – CE1.

The consequences are that every AI-assisted calculation step still requires manual data preparation upstream and manual transfer downstream, effectively inserting the tool into the middle of the manual workflow rather than replacing it. The cross-tool pattern is decisive for the readiness developed in Chapter 6. Among the first tier (Cost estimation) tools, no tool reaches L3 on Integration. The only L3 score for the entire benchmark belongs to Tool XD, a payment-term tool in the second tier. This means the integration capacity that exists at Company X is not transferred within the internal tools. The implication is that the integration constraint is not about tool design, but about the structural data infrastructure problem that must be addressed upstream of any individual cost estimation tool deployment.

5.4 Organization Dimension Findings

The Organization dimension pairs Use Cases & Value, the scope and demonstrable value of the AI tool, with the Workflow & Culture of the tool and its deployed team, which gives the overall picture of the human and process readiness of the company to adopt and sustain the tool in daily practice. The pairing is an analytical tool; a tool that delivers measurable value but cannot be embedded in the daily workflow and remains at the pilot stage, while a tool deeply embedded in the workflow but unable to demonstrate value loses its case for continued investment.

5.4.1 Use Cases & Value: The Scope Hierarchy

The most significant finding under Use Cases & Value is not which tool delivers the most value in absolute terms of cost estimation alone, but the consistent hierarchy of relevance that the cost engineers at Company X applied when evaluating tool utility. Tools directly serving the should-costing or price-estimation function were prioritized over tools addressing adjacent procurement functions, even when a few adjacent tools demonstrated higher operational maturity, since they provide the required value to cost engineers at Company X and the cost-calculation ideology. This concept is the empirical basis for the first and second-tier distinction used throughout this chapter.

Within the first tier, Tool B (Established – L3) currently delivers the strongest demonstrated value in the dataset. It combines should-costing for new parts, similar part matching from drawings and 3D models, raw material market linkage, and structured cost comparison in a single platform deployed at scale. Tool D (Established – L3) operationalizes image-based Printed Circuit Board (PCB) cost prediction by matching with existing library parts. Tool C (Pilot – L2) at Company C works on LLM-based software cost estimation, which reads the specification book requirements, delivering the most reliable timing for information sizing. Within Company X, Tool XA is the strongest first-tier candidate based on use-case alignment. The scope it addresses is precisely the priority that interviewees CE6 and ITH identified, but the pilot scope limits the current value. At the time of data collection, the pilot covered a limited number of parts as an initial rollout trial for the team to work in cost engineering and procurement. Tool XB (Pilot – L2) delivers more diverse value through designer-based costing during product development, with a short manufacturing lead time and lower overall product cost, but its use is structurally low because cost factors are not the designer's primary task.

Within the second tier, the strongest value evidence comes from Tool XC (Established – L3). Interviewee XC described the operational transformation in the following terms:

"What used to be half a working day every month, going through supplier news, financial signals, risk alerts manually, is now in one click, and the dashboard shows me everything. This time saving is real, and it is measured." – Interviewee XC.

The first-tier value for Tool C was derived from Interviewee C's statement:

"Before the tool, sizing a specification book took our team somewhere between eight and ten weeks. Now it is under one hour. That is not an incremental improvement; that completely changes how we can respond to customers and add value in our product." – Interviewee C.

The Tool XC and Tool C as first and second-tier value claims that demonstrate high, measurable operational outcomes and are achievable within the current group ecosystem. From a cost engineering perspective at Company X, the consistent focus is on tools that directly serve their accountable function, which is why Tool C is highly valued in software

cost sizing and elevated to first-tier status as a direct cost-estimation support tool at a peer organization. This is also a part of solving the challenge that companies face during AI use case selection, and it is evident that Company X resolves this through a practical approach of scope applied by an experienced domain expert, that is, the cost engineers prioritizing tools that directly support their accountable function, regardless of how mature adjacent tools are in place.

5.4.2 Workflow & Culture: The Pilot Ceiling and Calibrated Trust

The Workflow & Culture findings present the most consistent pattern in the entire benchmark: nine out of ten tools sit at L2 (Pilot), and only Tool B reaches L3 (Established) through its multi-unit rollout, role-based access governance, and established licensing model. This near uniformity is itself an analytical finding. The boundary between pilot adoption and integrated daily practice is not a tool-specific problem but a systematic challenge in automotive AI adoption in cost engineering that appears regardless of tool complexity, company size, and investment.

The Company X internal interviews provide the deepest account of what sustains this pilot structure. Two mechanisms dominate: the first is the long tenure with the same cost calculation software in the cost engineering team, which produces a deeply habituated way of working that AI tools disrupt cognitively as well as operationally. CE2 articulated the nature of this challenge with precision.

“The challenge is not the technology. The challenge is that people have been using legacy cost calculation software for fifteen years. It is their language. Asking them to trust an AI recommendation on top of that is a very different kind of change from just updating the software.” – CE2.

The second mechanism is the black-box skepticism that the cost engineers articulate, a concern that is not irrational but a professional accountability pressure. Cost engineers must justify their outputs in supplier negotiations. A tool that produces a result without auditable reasoning exposes the cost engineers to challenge their work.

“Black-box skepticism: people are afraid of AI. It’s really a cost calculation. The split and the machines behind this would be helpful.” – CE3.

The theoretical implication is specific: the barriers are part of the corporate culture in large organizations and act as obstacles to AI adoption. The Company X evidence adds that, specifically, this cultural barrier in cost engineering is not generally about new technology adoption but domain-specific accountability pressure. The implication of the company design is that generic change management training will not completely resolve the pilot structure. What is required is an approach to developing a tool with calibrated trust that enables designated sanity checks, transparent reasoning protocols, and human-in-the-loop

verification, and positions AI output as an informed starting point rather than a final answer. CE4 captured this as:

"It would be good if it provides everything OK, and you just do the sanity check. OK, makes sense. I can go into a negotiation with those numbers." – CE4.

Several cost engineers independently converged on this framing, suggesting it is an emergent organizational norm rather than a research criterion. The inductive coding of the cost engineer's interviews surfaces a complementary theme that does not fit any single benchmark factor but is relevant to the Organization dimension: a generational and skill divide in AI proficiency. CE6 noted that prompting skills vary widely across staff, with younger staff entering university-level AI literacy and long-tenured staff with minimal exposure. This divide creates internal separation in how the same tool is used and has implications for the upskilling agenda that supports the move from L2 to L3. It is documented as an inductive observation rather than a benchmark factor and is discussed further in Chapter 6 in connection with the practical implications for the adoption of sequencing.

5.5 Environment Dimension Findings

The Environment dimension pairs External Drivers, such as regulation, competitive pressure, and strategic forces, shaping the adoption urgency with the Ecosystem Readiness factor, consisting of the data infrastructure, supplier transparency, and partnership context, from a broad perspective that determines whether an AI tool can operate on the data it requires.

5.5.1 External Drivers: Named Strategy as Differentiator

External Drivers scores range across the benchmark from L1 (general competitive pressure only) to L3 (proactive, strategy-linked deployment). The pattern that emerges is that tools that are anchored to a named organization-wide strategy or a specific external force that is driven by a regulatory mandate, competitive market positioning, or an efficiency target score higher than the tools deployed in response only to general competitive pressure. Tool B is the clearest example; its rollout is part of the centralized cost estimation initiative explicitly intended to align AI investment use cases. This strategic anchoring is reflected directly in the L3 score and equally in the tool's mature workflow integration. Tool D is similarly anchored through an efficiency and headcount strategy, and Tool C through a market positioning logic about "China speed" and customer experience competitiveness.

Within Company X, the strategic anchoring is uneven. Tool XA scores L3 due to the digital transformation strategy and operational deployment driver to increase efficiency, time saving, and cost reduction, which is explicitly cited by the interviewee. Tool XE also reaches L3 through this clear logic. Tools XB, XC, and XF, however, score L1 not because they lack strategy, but the intent is articulated in general competitiveness without anchoring to a named target. Interviewee ITH framed Company X's overall position with a useful external benchmark:

“We are early movers. Company X is recognized externally as an early enterprise adopter of AI. That is a positioning. The company sees AI as strategic, but translating that into which tool, for which team, at what maturity level, that is the work that still needs to happen.” – ITH.

The L3 anchoring exists at the Company X level; the unevenness is in how individual tools are linked to it. The cost engineer interviews also introduce a complicating dimension that the four-level External Drivers factor does not fully capture, which is the regulatory variety. CE1 described a concrete instance where regulatory pressure reversed the adoption dynamic entirely:

“There was a point where the U.S team just stopped. Compliance said no AI tools until we have clarity on the regulations. Everything went on hold. That tells you it is not just about having the technology. The regulatory environment shapes what you can use, and where.” – CE1

This illustrates that external drivers can simultaneously accelerate and constrain adoption depending on jurisdiction, and it is a broad discussion in Chapter 6 as a boundary condition on the recommendation logic.

5.5.2 Ecosystem Readiness: The Structural Bottleneck

Ecosystem Readiness is the most uniformly constrained factor in the entire benchmark. Eight of the ten tools score L1 (Ad-hoc), the lowest level possible. Only Tool XC and Tool XD reach L2, and no tools reach L3 in the ecosystem. This finding is decisive for the readiness-reading perspective. It means that the prerequisites across the group ecosystem, namely, the data preconditions for most AI tools in cost engineering, are not yet in place or even considered. Tools that depend on structured, standardized, or supplier-provided cost data, which applies to most ML-based cost estimation tools, operate in an ecosystem that cannot yet scale or requires constant manual data uploads for system updates.

The cost engineer interviews provide a precise account of why this matters in practice. Supplier-provided cost breakdowns arrive in varying formats and with varying levels of required commercial information. CE5 described suppliers as practicing strategic data minimization, a precisely chosen phrase that captures the competitive strategy behind the data quality problem experienced by the Company X cost engineering team. Interviewee B, with the broadest current deployment in the dataset, acknowledged the same constraint in concrete terms:

“You can validate everything strictly and end up with a beautiful database that covers thirty percent of your parts. Or you use everything and get broad coverage, but you also have noise that the model learns from. There is no perfect answer to that trade-off until the ecosystem-level data quality improves.” – Interviewee B

Interviewee D also positions the constraint at the supplier relationship level:

“I’m not sure that we will have such partnerships with our suppliers where we can share all the information we need. Our organization didn’t know how powerful a partnership could be in such an environment with our suppliers.” – Interviewee D

Interviewee CE2 articulated an aspirational counter-position that frames the gap as an institutional rather than a technical problem, in which shared data ownership requires mutual consideration between supplier and customer, with structured agreement on third-party use. Interviewee XA identified the same constraint on the supplier side, noting that Tool XA is not heavily engaged in supplier ecosystem integration, meaning that even when the technology platform is developed, the supplier-side data flow is not designed or initiated efficiently. Two further inductive observations emerged from interviewees CE6 and ITH, respectively. Interviewee CE6 highlighted the compounding effect of internal data fragmentation at Company X:

“We have two separate data flows, one for serious production and one for spare parts. The part numbers, the structures, the cost logic, and everything are different. Any AI tool that needs historical data must deal with the split from the beginning.” – CE6.

Interviewee ITH confirmed that this internal data fragmentation runs alongside the external supplier-side data gaps to be addressed. The TOE framework analysis positions Ecosystem Readiness as the single largest structural restraint to AI adoption in cost engineering at Company X and across the studied peer organizations. Unlike the Technical Scope or Workflow & Culture, which are constraints that internal decisions can address, Ecosystem Readiness is fundamentally a relational problem requiring coordination with external parties. This does not make it intractable, but it does mean that tool selection decisions must account for the ecosystem constraint.

5.6 Cross-Dimension Synthesis

Reading the findings across all six factors and across all ten tools identifies three analytical patterns. These patterns go beyond the individual factor findings and constitute the analytical contribution of this chapter to the research questions. Figure 3 below shows the visual mapping of Tool Levels across all six factors of the TOE dimension to better understand the pattern discussion.

Pattern 1 – The Pilot Ceiling is a shared Industry Condition. Nine out of ten tools sit at L1 or L2 across Workflow & Culture and Integration, with the single exception of Tool B at L3. No first-tier (cost estimation) tool in the dataset has reached embedded daily workflow status. This is not a coincidence of sample sets but reflects a structural characteristic of AI adoption in cost engineering at this moment in the automotive OEM context. The pilot ceiling is a shared condition rather than Company X's specific deficit, which has direct implications for

how the readiness signals are interpreted in Chapter 6. The absolute readiness gap between Company X and its peers is smaller than a tool-by-tool comparison might suggest, and the question is which company will cross the pilot-to-integrated boundary in the first place.

Pattern 2 – The Data-Tool Mismatch Condition. The tools with the most architectural complexity in Technical Scope (Tool XA, Tool C – LLM-based, Tool XF – GenAI-based) score the lowest on Ecosystem Readiness (L1). Conversely, the tool with the most mature ecosystem integration scoring (Tool XC, with multiple internal system connections) has lower technical complexity and addresses a non-direct cost estimation use case. This mismatch means that advancing on one dimension often comes at the cost of another. In particular, the cost estimation tools at both internal Company X tools and the first-tier peer tools, including Tool C, are precisely the ones where supplier and project-side data is least standardized and structured. This pattern suggests that a sequenced adoption strategy that builds ecosystem readiness first and then deploys more complex cost estimation tools, which is more realistic for Company X than parallel advancement across all six factors. The recommendation matrix in Chapter 6 is built around this sequencing logic.

Pattern 3 – Strategic Anchoring as an Internal Variable Condition. Across the dimensions, the variable that is under control within each organization is the External Drivers. The technical scope is shaped by either external consultants or internal software team developers. Integration depth depends on legacy data architecture; Workflow & Culture is shaped by long-tenured organizational habits. Ecosystem readiness depends on external parties. Only the External Drivers are the exception, fundamentally controlled by the internal positioning choice of whether to be anchored to a deployment with a named organization-wide strategy or to leave it as a general competitive response. The tools at peer organizations that have moved furthest (Tool B, Tool D) are precisely those with the strongest strategic anchoring. The recommendation logic in Chapter 6, therefore, identifies External Drivers as the leverage point with the highest internal feasibility for Company X.

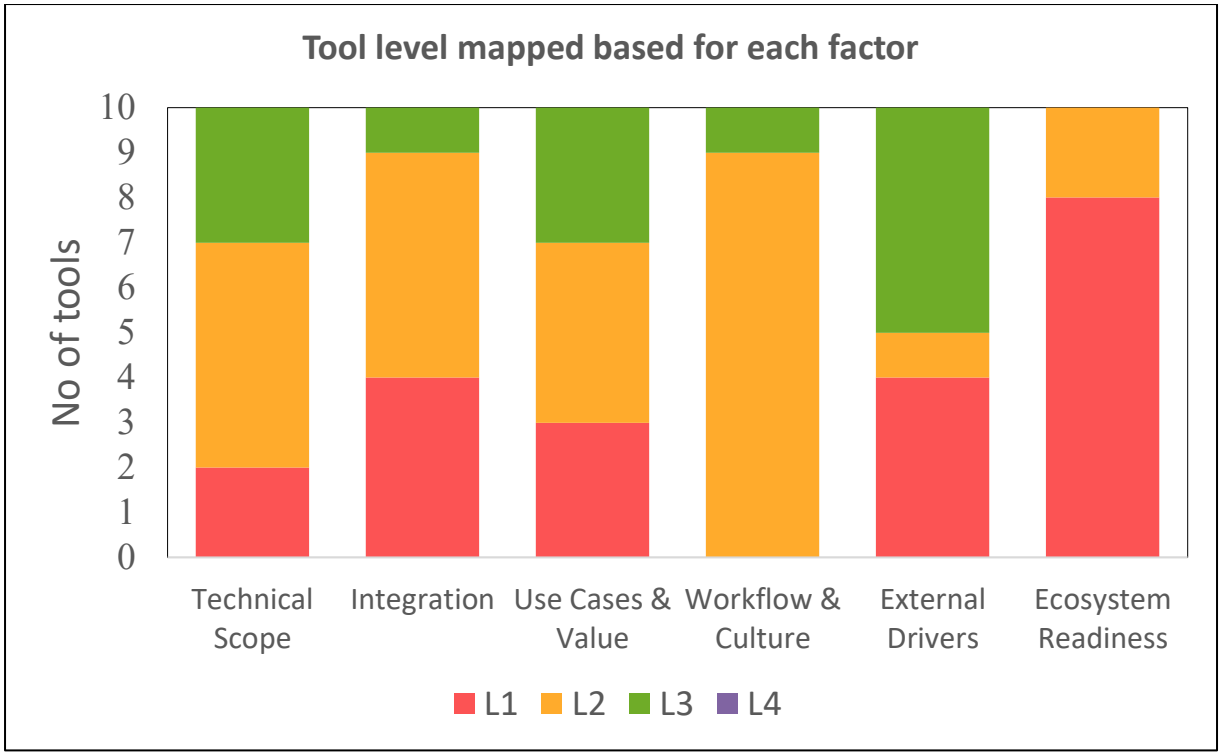


Figure 3. Bar graph mapping Tool Levels on each factor of the TOE dimension (Source: Authors)

5.7 Benchmark Rankings

Table 7 consolidates the level assignments across the six factors and the resulting weighted scores for each of the ten benchmarked tools. The full benchmark table comprises 60 keyword–level pairs (six factors across ten tools) and 10 total scores that drive the ranking column. The total score reflects the sum of weighted scores across the six factors, where Technical Scope uses reverse weighting (L1 = 4, L2 = 3, L3 = 2, L4 = 1) and the remaining five factors use forward weighting (L1 = 1, L2 = 2, L3 = 3, L4 = 4). The ranking column orders tools by total score with a tiebreaker applied through the Company X requirement and scope described in section 5.4.1, which gives priority to first-tier (Cost estimation) tools where total scores are tied. The full coding workbook with quotes, keywords, and factor justifications is provided in Appendix F.

Three observations from Table 7 inform the analysis in Chapter 6. First, no tool achieves a total score higher than 16 (Tool B), and the maximum theoretical score of 24 is not approached by any tool in the dataset. The benchmark, therefore, captures an industry-wide transitional state rather than a fully mature one. Second, the score range is compressed to 9-16 across ten tools, meaning tool selection cannot rely on absolute total scores alone. Each factor profile carries a different analytical weight for every tool. Third, when the requirement from Company X's priority is applied as the tiebreaker, the ranking reorders the second-tier high scorers below their numerical position. This is also a part of the Use Cases & Value factor-specific need and implementation logic for the cost engineering team of Company X. Tool XD, with a total score of 14, sits at position 7, below first-tier tools with scores of 12

and 11, because its use case is procurement supporting rather than direct cost estimation purposes. This is an analytical choice that reflects the logic of Section 5.4.1 and the case selection concept of Chapter 4. The rank order in Table 7 is consistent with the sequence established in the full coding workbook (Appendix F), which documents the factor justifications and keyword-to-level assignments underlying each score. The implications of this ranking for prioritized AI adoption decisions at Company X are developed in Chapter 6. The total score functions in this study as a structured navigation aid rather than as a measurement of absolute tool performance. Since the primary purpose at Company X is to maximize the use of cost-estimation-related AI tools, the ranking is designed to compare cost-estimation tools first and to position procurement-support tools secondarily on the same factor set, which lets the reader see what those tools additionally offer without losing the cost-estimation focus.

Tier	Rank	AI Tool	Technology (T)		Organization (O)		Environment (E)		Scoring
			Technical Scope	Integration	Use Cases & Value	Workflow & Culture	External Drivers	Ecosystem Readiness	
1st	1	Tool B	Algorithm	Live	Compare CBD	Standard with restriction	Strategy	Not considered	16
			L1	L2	L3	L3	L3	L1	
	2	Tool D	Rule	Limited access	Cost Prediction	Limited adoption	Strategy	No data transparency	13
			L2	L1	L3	L2	L3	L1	
	3	Tool XA	LLM	Manual data	Cost estimation	Change initiated	Digital Transformation	Not considered	12
			L3	L2	L2	L2	L3	L1	
	4	Tool XB	Algorithm	Limited access	Cost estimation	Pilot	Competitive	Not considered	11
			L1	L1	L2	L2	L1	L1	
	5	Tool C	LLM	Deployment pending	Spec book sizing	Change initiated	Cost, Strategy	Limited access	11
			L3	L1	L2	L2	L3	L1	
	6	Tool A	Gen AI	Stand alone	Gathering cost data	Change initiated	Competitive	Not considered	9
			L2	L1	L1	L2	L1	L1	
2nd	7	Tool XD	Gen AI	Connected	Gathering data	Pilot	Regulation	Not considered	14
			L2	L3	L2	L2	L2	L2	
	8	Tool XC	Multiple	Legacy systems	Risk analysis	Change initiated	Competitive	Internal data dependency	13
			L2	L2	L3	L2	L1	L2	
	9	Tool XE	Gen AI	Legacy systems	Payment tracking	Pilot	Digital Transformation	Not considered	12
			L2	L2	L1	L2	L3	L1	
	10	Tool XF	Multiple	Legacy systems	Gathering data	Pilot	Competitive	Not considered	9
			L3	L2	L1	L2	L1	L1	

Table 7. Tool score and ranking with Keywords and Level definitions (Source: Authors)

6. Discussion

6.1 Introduction

This chapter discusses the empirical findings presented in Chapter 5 through the lens of the integrated TOE benchmark framework defined in Chapter 3 and against the literature reviewed in Chapter 2. The aim is to interpret what the findings mean for the three research questions and, beyond them, what the study contributes to practice and to the broader ethical and societal questions surrounding AI adoption in cost engineering. Section 6.2 addresses RQ1 by interpreting the AI tool landscape across the studied OEMs. Section 6.3 addresses RQ2 by examining what the integrated TOE benchmark reveals that conventional approaches do not capture. Section 6.4 addresses RQ3 by translating the benchmark scores into prioritized recommendations for Company X. Section 6.5 discusses the broader practical and ethical implications arising from the analysis.

6.2 The AI Tool Landscape Across the Group (RQ1)

The findings in Chapter 5 show that AI tools across Company X and its four peer organizations cluster into two analytical tiers with respect to the cost engineering scope. The first-tier tools (Tool A, Tool B, Tool C, Tool D, Tool XA, Tool XB) directly address should-costing, parametric estimation, LLM-based software cost sizing, image-based PCB costing, and LLM-based price reasoning. The second-tier tools (Tool XC, Tool XD, Tool XE, Tool XF) address procurement support functions, supplier risk profile, payment terms, document retrieval, and control tower, which contribute to cost engineering indirectly. This tier distinction is not a metaphor but an empirically grounded scope distinction that the cost engineers at Company X applied consistently when assessing the tool relevance.

The literature anticipates this growth of AI use cases. Balkan and Akyuz (2025, p.2) observe that AI tools span a wide variety of procurement use cases and that the practical challenge for organizations is not the absence of AI but the abundance of overlapping options without a structured way to compare them. Within the company alone, six AI tools at varying maturity levels operate on different use cases. The integrated TOE benchmark addresses this selection challenge by producing a structured comparison that scopes both relevance (the tier distinction) and recommendation (the level assignments) in a single framework methodology. In terms of demonstrated benefits, the cost estimation tools at peer organizations have begun to deliver measurable outcomes. Tool B operationalizes should-costing through similar part matching across drawings, 3D models, and CBDs. Tool D operationalizes image-based PCB cost prediction with claim management benefits and cycle time reduction. These observations are consistent with Bodendorf, Merkl and Franke (2021), who confirm that ML is increasingly used for predictive cost estimation in supply management, and with Hammann (2024) on big data and ML in automotive cost estimation. The cost estimation tools at Company X have not yet reached this level of demonstrated benefit, which aligns with the functional-level finding by Bhogal and Rudqvist (2025) that Company X is in active transition, with foundational readiness elements still developing.

Three conditions emerge consistently as preconditions for first-tier tools to deliver value in practice. Data quality and supplier-side data availability are critical, as Bodendorf and Franke (2024, p.3) also argue. Workflow ownership and change management matter, since organizational embedding is critical for AI value realization in procurement (Spreitzenbarth, Bode and Stuckenschmidt, 2024, p.1). Architectural complexity affects adoption speed inversely — simpler tools deploy faster while more advanced tools remain in pilot — which differs slightly from parts of the AI in procurement literature that emphasize the value of advanced architectures. The empirical answer to RQ1 is therefore that AI tools are present across cost engineering and procurement in the studied group, that demonstrated benefits cluster in first-tier cost estimation tools at peer organizations further along in deployment than Company X, and that the organizational and environmental conditions for value realization are dominated by data, workflow, and architectural fit rather than by raw technology advancement or complexity.

A broader implication of this tier distinction for AI in procurement literature is that the field has started to discuss AI adoption as if the procurement function were one large function. Studies addressing AI tools in procurement, including those studies in this thesis review, typically treat procurement as a unified domain of intervention, aggregating use cases from spend analytics to contract management to supplier onboarding without disaggregating the sub-functional readiness profile that distinguishes them. The Company X evidence challenges this assumption directly. Cost engineering, a support function and sub-function responsible for should-costing, supplier negotiation preparation, and CBD analysis, operates under a set of constraints that are distinct from those of general procurement. It requires auditability, defensibility under supplier challenge, and integration with engineering data systems that do not share the data architectures of procurement platforms. The implication for future research is that AI adoption studied in cost engineering and procurement should disaggregate the function into sub-functions with distinct readiness or recommended profiles, rather than treating procurement-wide AI adoption as a unified analytical object. The tier distinction developed in this study is an empirically grounded starting point for that disaggregation.

Research that looks only at the capabilities of AI tools and their use cases does not by itself serve individual companies trying to understand AI tool capabilities; it gives only a development direction without practical grounding. The selection criteria for AI tool adoption should include internal assessment, capabilities, and the best use case with minimal complexity for the existing system. Since data on AI tools is fragmented with respect to the cost engineering and procurement function, it is difficult to transfer knowledge and decision-level priorities to a practical organization that is in the initial stage of AI tool adoption.

6.3 What the Integrated TOE Benchmark Reveals (RQ2)

The integrated TOE benchmark with six measurable factors enables a structured cross-tool comparison that produces the three patterns described in Section 5.6: the pilot ceiling as a shared industry condition, the data-tool mismatch, and strategic anchoring as the dominant

internal variable. Read together, these patterns show that the framework is not producing a flat ranking but a structured readiness reading in which each TOE dimension contributes a different kind of signal. This section discusses what the reading reveals that a tool-by-tool ranking result would not provide.

The first analytical insight that the integrated framework surfaces is the value of the reverse scoring logic on Technical Scope. Without it, the most architecturally advanced tools would rank highest despite being the least adopted, which would invert the empirical reality of cost engineering practice at Company X. The reverse scoring logic encodes the convergent evidence from five of six cost engineers and the IT head that complexity is an adoption barrier rather than an indicator of capability. This is a context-specific weighting, supported by triangulation that conventional maturity models such as AIMM, AI-CAM, CMM, and CRISP-DM would not generate, since they treat high Technical Scope as an indicator of the capability of the tool's measurable outcome. The integrated framework is therefore a more trustworthy structure for AI adoption decisions in cost engineering at this transitional stage, where the company's absorptive capacity has not yet reached a technical frontier level.

The second insight is that benchmark scores can be read as recommendation signals through the dimensional structure rather than as raw performance scores. Where Tool B at Company B reaches L3 on Workflow & Culture but Tool XA at Company X sits at L2 on the same factor, the gap is itself a signal that Company X is one step behind on workflow integration for first-tier tools. This creates an opportunity for actionable recommendations that a separate assessment framework would not contribute to, because it identifies the specific factor on the tool-level where the gap exists. The framework is therefore a more practical instrument than a dual benchmarking and assessment design that Flechsig, Anslinger, and Lasch (2022), Riti, Bacali and Abrudan (2025), and Marhraoui (2026) implicitly assume by treating benchmarking and readiness assessment as separate exercises. The integrated design eliminates the conceptual duplication and the risk of conflicting scoring schemes that separate frameworks inevitably create.

The third insight is the cross-dimensional reading pattern that single-dimensional reading cannot generate. The pilot ceiling is visible only because Workflow & Culture and Integration are read together. The data-tool mismatch is visible only because Technical Scope and Ecosystem Readiness are read together. The strategic anchoring pattern is visible only because External Drivers is read against the absolute scores on the other five factors. Each of these patterns becomes a readiness assessment signal. The pilot ceiling tells Company X that workflow integration is the binding industry-wide constraint, the data tool mismatch tells Company X that ecosystem readiness must be put in place before architectural advancement, and the strategic anchoring pattern tells Company X that named strategy linkage is the highest feasibility of internal action. The empirical answer to RQ2 is therefore that AI tools across Company X and its peer organizations can be benchmarked using the integrated TOE benchmark dimensions to produce both a comparative ranking (Table 7) and a structural readiness assessment reading, which comprises the three patterns mentioned in Section 5.6, within a single analytical exercise.

The methodological argument for integration over separation carries implications that extend beyond this thesis study. In industrial AI governance, the practice of running separate benchmarking and readiness assessments is common, but commissioning capability reviews alongside maturity reviews produces two sets of scores that must then be manually reconciled. The tension this creates is real: a highly capable tool with low organizational readiness, or a well-embedded tool with limited technical capability, represents precisely the kind of trade-off that the integrated TOE benchmark is designed to make visible within a single instrument. The finding in this study that the most technically capable tools score lowest on Ecosystem Readiness is not a paradox but a structural insight, which a capability assessment would make invisible since it would score the LLM-based tools at the top and the rule-based tools at the bottom without surfacing the data infrastructure bottleneck that governs actual adoption feasibility. By encoding that mismatch within a single scoring scheme, the integrated framework lets decision makers see the sequencing logic directly, where a separate framework approach would require additional analytical steps. For other industrial organizations designing AI governance instruments, this argues for treating benchmarking and readiness assessment not as parallel tracks but as dimensions of the same analytical exercise from the start.

6.4 Prioritized Recommendations for Company X (RQ3)

This section translates the benchmark scores into a prioritized AI tool adoption decision for Company X. The translation logic is that each tool's score profile reveals the binding constraint factor, the single factor whose level is most limiting further adoption progress. The recommendation for that tool then targets the binding constraint specifically, rather than recommending a generic improvement across all factors. The benchmark ranking in Table 8 provides the priority sequence, the order in which Company X should engage with the ten tools, and reflects the case scope of maximizing cost-estimation tools first while keeping the procurement-support tools visible on the same factor set. The recommendation logic itself is binding-constraint driven and therefore differs from a simple "act in rank order" criterion, since two tools with the same total score may face different binding constraints and call for different actions.

6.4.1 First Tier Recommendation Logic

Reading the first tier (cost estimation) findings produces six observations that are relevant to Company X, presented here in two groups: peer organizations' tools in rank order of their benchmark score (Tool B, Tool D, Tool C, Tool A), followed by internal Company X tools (Tool XA, Tool XB). This grouping does not deviate from the ranking order; it is a reminder that external tools require engagement decisions that set the boundary conditions, while internal tools are directly influenced by Company X.

First, Tool B (Rank 1) is a peer organization tool, and the only first-tier tool in the dataset that has crossed the pilot ceiling by reaching L3 on both Use Cases & Value and Workflow &

Culture. The recommendation is to engage with Company B's rollout program to evaluate licensing terms and the role-based access for Company X's cost engineering scope. Second, Tool D (Rank 2) carries the second strongest adoption signal in the first-tier dataset, scoring L3 on both Use Cases & Value and External Drivers. Its PCB-image capture and price prediction use case is directly relevant to cost estimation, but its applicability depends on whether Company X's commodity portfolio includes electronics-intensive requirements. The recommendation is therefore to assess fit and, if needed, adopt the tool from the peer organizations. Third, Tool XA (Rank 3) is Company X's strongest internal first-tier tool based on use case alignment. It can realistically progress from L2 to L3 on Workflow & Culture, and the pilot has begun, which demonstrates that the cost engineering team will be allocated to a tool usage trial. The gap for Tool XA is about the deployment and not the technical capability; therefore, the rollout should be planned from the limited pilot to wider use with the sanity checks and trust protocols mentioned by interviewees CE3 and CE4.

Fourth, Tool XB (Rank 4) sits at L1 on most factors, and the structural limitation is that designers do not engage with cost analysis as a primary task. The recommendation is to keep the tool simple, expand the user base to cost engineering, and not to invest in deep integration or further investment, since the use case does not completely support cost engineers. Fifth, Tool C (Rank 5) is a first-tier peer organization tool that has not yet been deployed and is pending board approval at Company C. The constraint is about the governance delay, and not the technical or use case gap. The recommendation is to monitor Company C's rollout outcome, and since it's an LLM-based software cost sizing approach, it is relevant to future Company X initiatives to automate the cost engineering workspace. Sixth, Tool A (Rank 6) is the only first-tier tool for which no adoption recommendation is made at this stage. Its Use Cases & Value score sits at L1 with no demonstrated cost engineering-specific outcome; no further action is needed beyond monitoring until Company A shows evidence for measurable improvement.

6.4.2 Second Tier Recommendation Logic and the Ecosystem Gap

The four procurement support tools at Company X carry different recommendation logic. Tool XD (Rank 7) reaches L3 on Integration, the highest integration score in the dataset, and is a candidate for sustained operational use and as an architecture reference for Tool XA integration planning. Tool XC (Rank 8) is mature on Use Cases & Value (L3), which clearly shows measurable benefits and is a strong candidate for continued investment with improvement in supplier data quality agreements that would advance its Ecosystem Readiness from L2 to L3. The binding constraint for Tool XC is the unanchored External Drivers factor, and advancing the tool toward the digital transformation strategy or a named cost engineering program would be most beneficial. Tool XE (Rank 9) and Tool XF (Rank 10) both sit at L1 on Use Cases & Value, which means the binding constraint is the absence of a measurable value claim. The recommendation is to hold off on approaching the tool for any immediate deployment before acting on the tier-one and tier-two priority tools.

Beyond the tool-level recommendations, one strategic-level intervention emerges from the cross-tool reading. The Ecosystem Readiness gap (eight of ten tools at L1) is not solvable at the tool-level. It requires group-level coordination on supplier-side data standardization, supplier tier-1 partnership protocols, and structured CBDs of data exchange formats. This is upstream of all first-tier cost estimation tools at Company X. Without this strategic intervention, no tool-level investment in cost estimation will reach its full value and will be limited to internal Company X or peer organization data without supplier-side enrichment. There is little prior discussion or empirical evidence in the literature on the Ecosystem Gap, since this is an early-stage debate: most AI tools are currently operating in the pilot ceiling phase, where data is fed manually or where minimal integration with legacy systems exists even within the same organization.

6.4.3 Convergence with Previous Study at Company X

The pre-study by Bhogal and Rudqvist (2025) at Company X provides external validation for the recommendation logic developed here. They found Company X to be mature in education, developing in data quality, leadership commitment, and cross-functional collaboration, and emerging in data infrastructure, change management, and organizational adaptability. The functional level readings converge with the tool-level benchmark on three points. Both find change management at the developing stage; the benchmark shows Company X tools at L2 on Workflow & Culture, while pre-study categories, such as change management, are at the emerging level. Both identify data infrastructure as a constraint; the benchmark shows weak integration scores across first-tier tools at Company X, and pre-study categorizes infrastructure at the emerging level. Both find strategic alignment as partial; the benchmark shows mixed External Drivers scores at Company X compared to the more uniformly anchored peer organizations of Company X, while the pre-study notes leadership commitment at a developing rather than a mature level. The convergence is methodologically important because it shows that the recommendation priority is consistent regardless of whether it is measured at the functional or tool-level, which strengthens the study's validity and addresses RQ3 with strong practical working data.

Table 8 presents the practical recommendation logic across all the tools and the one strategic-level intervention. The table is read left-to-right, with the ranked position of AI tools, the constraint factor that needs intervention, the action that targets the constraint, and the type of action it represents. The empirical answer to RQ3 is therefore that the integrated TOE benchmark framework scores can be translated into prioritized AI tool adoption decisions for Company X by reading each tool’s score profile to identify its constraint and targeting the recommendation specifically at that constraint. The result is not a single recommendation but a structured set of actions, ordered and driven by feasibility and tier priority.

Rank	Tier	Tool	Constraint Factor	Recommended Action	Result
#1	**1st**	**Tool B**	Already at L3 on Workflow & Culture	Engage Company B for licensing terms evaluate role-based access for Company X scope	**Adopt from peer**
#2		**Tool D**	Scope relevance to Company X - Commodity specific requirement	Assess PCB-image scope applicability before any adoption decision	**Assess and adopt from peer**
#3		**Tool XA**	Workflow & Culture at L2	Accelerate pilot rollout; build sanity-check protocol; assign tool champion	**Scale now**
#4		**Tool XB**	Use frequency low; designer-led	Maintain current scope; do not over-invest in integration	**Maintain scope**
#5		**Tool C**	Board approval pending; governance delay not technical gap (Integration at L1, Workflow at L2)	Monitor Company C board approval and deployment; evaluate LLM software cost sizing approach for relevance to Company X software cost engineering scope	**Monitor for deployment and adopt**
#6		**Tool A**	Use Cases & Value at L1; no additional AI related cost-estimation value	Monitor until Company A demonstrates measurable cost-engineering outcome	**Monitor only & adopt only if use case improved**
#7	**2nd**	**Tool XD**	Use case is second tier	Sustain; use L3 integration architecture as reference for Tool XA planning	**Sustain**
#8		**Tool XC**	External Drivers at L1 (no named anchor)	Anchor to digital transformation strategy; formalize supplier data-quality agreements	**Continue and deepen**
#9		**Tool XE**	Use Cases & Value at L1	Hold adoption decision until it is deployed and based on team requirement adopt	**Hold adoption decision**
#10		**Tool XF**	Use Cases & Value at L1 External Drivers at L1	Hold adoption decision until it is fully developed and based on use case & team requirement adopt	**Hold adoption decision**
Strategic-level intervention **Ecosystem Readiness gap (8/10 tools at L1)**				Group-level coordination on supplier data standardization; Tier-1 partnership protocols; structured cost breakdown exchange formats	**Group-level mandate**

Table 8. Prioritized recommendation matrix mapping each tool to an action for Company X (Source: Authors)

6.5 Broader Implications

6.5.1 Practical Implication – Sequenced Adoption Rather Than Parallel Advancement

The data-tool mismatch surfaced in Pattern 2 of Section 5.6 carries a direct practical implication that extends beyond Company X. Organizations facing simultaneous constraints across Technical Scope, Integration, and Ecosystem Readiness cannot advance all three in parallel, because progress on the most complex tools is conditional on data infrastructure that does not yet exist at the ecosystem level. A sequenced strategy that first strengthens ecosystem readiness through supplier-side data agreements, then improves integration through architectural investment, and only then deploys more complex cost estimation tools, is the realistic pathway forward. For Company X specifically, the sequence to accelerate Tool XA from L2 to L3 on Workflow & Culture using the embedded cost engineering team as the primary deployment platform, maintain Tool XB at its current scope without over-investment, initiate strategic supplier-side data standardization as a group-level intervention, and anchor each subsequent tool rollout to a named strategic driver to avoid the L1 external drivers constraint observed in Tools XB, XC, and XF. The same logic can be generalized to other automotive OEMs at similar transition points, since the patterns observed in this study are not specific to Company X but shared across the peer organizations studied.

The sequencing logic above translates into differentiated recommendations by the stakeholder group. For cost engineering managers at Company X, the highest priority action is to formalize the Tool XA sanity check protocol and assign a named tool champion within the cost engineering team. Without this ownership, the pilot will stall at L2 regardless of technical readiness. For the IT function, the most appropriate near-term investment is not in the AI layer itself but in the data infrastructure that connects legacy systems to upstream and downstream, since the integration scores show this to be the major constraint across all first-tier tools. For senior leadership and group procurement strategy, the decisive action is to initiate a supplier data standardization program at the group level. This is not a tool decision, but a commercial and governance decision that creates the ecosystem preconditions for every subsequent cost estimation tool investment. For peer organizations of Company X benchmarked in this thesis study, the findings confirm that the pilot ceiling is a shared condition; the same condition is identified in every research article that is studied, where every case study deals with AI tools in the pilot stage, trying to improve the overall outcome or profit criteria. The organization that first operationalizes calibrated trust at scale through transparent reasoning protocols and designated verification roles will establish the benchmark for the tool's highest value that others are measured against.

6.5.2 Ethical and Societal Implications – Trust, Negotiation Power, and Supplier Relations

The convergent finding that cost engineers articulate is calibrated trust, transparent reasoning protocols, designated sanity check roles, and human-in-the-loop verification as the

operational condition for AI adoption; these raise an ethical dimension that extends beyond the organizational assessment. Cost estimation outputs anchor supplier negotiations and AI-supported cost estimation, therefore shifting the informational balance of those negotiations. Three ethical questions follow: First, when an organization can generate AI-assisted should-cost models while a supplier cannot, what governance ensures that the resulting commercial advantage remains within the bounds of fair negotiation rather than becoming a form of pressure on the supplier? Second, when supplier data is required as input to the AI model, such as raw material costs, process information, and capacity data, what frameworks ensure that commercially sensitive information is handled with appropriate confidentiality and not repurposed beyond its consented use? Third, what explainability obligations does the organization take on when it deploys an AI-assisted cost number in negotiation without the supplier's involvement in the produced number? These questions are not resolved by the framework, but they are raised by the interview findings. The literature on AI in procurement (Spreitzenbarth, Bode and Stuckenschmidt, 2024) and on AI ethics in industrial contexts (Riti, Bacali and Abrudan, 2025) has begun to articulate them. This thesis surfaces them as live operational concerns at Company X and identifies them as topics the AI governance discussion must address as deployment matures from pilot to integrated practice.

6.5.3 Policy and Industry Implications

The findings carry implications that reach beyond Company X and its peer organizations to the industry and policy levels. At the industry level, the pilot ceiling observed uniformly across all ten tools suggests that the automotive OEM sector is collectively positioned in an early majority adoption phase for AI in the cost engineering function, past the early adopter stage but not yet at the integrated practice threshold that would constitute the mainstream adoption level. This positioning has consequences for how industry bodies, standard organizations, and technology vendors understand their respective roles. Automotive industry associations could accelerate the crossing of the pilot ceiling by establishing common data exchange standards for CBD formats, reducing the ecosystem standardization burden that individual organizations cannot solve in isolation. A clear case example is this study, where the near-universal L1 Ecosystem Readiness score gives a measurable picture of the gap. Vendors of advanced technologies such as LLM-based cost estimation face a parallel design challenge: the market for more complex architectures is limited not by demand but by the absorptive capacity of professional cost engineers, who put auditability ahead of raw capability. Organizations that invest in explainability layers and transparent reasoning will be better positioned to cross the calibrated trust threshold that this study identifies as the critical adoption precondition.

At the policy level, the EU AI Act does not distinguish between AI in internal cost modeling, where outputs inform human judgment, and AI in supplier-facing negotiations, where fragmented information creates a fundamentally different governance challenge. The findings of this study suggest that this distinction matters and that the cost engineering context makes it especially visible. An AI-assisted cost model used internally as a starting point for a human-reviewed negotiation position poses a different governance challenge than one

deployed as the direct basis for a supplier price challenge without human review or without disclosure of the reasoning chain. Policymakers developing sector-specific guidance on AI in Business-to-Business (B2B) commercial contexts would benefit from the kind of functional-level granularity this study provides: the recognition that within a single procurement function, different sub-functions operate under different informational dynamics and therefore require differentiated governance frameworks. This study does not resolve these policy questions, but it generates the empirical evidence that makes them legible as concrete operational concerns rather than abstract regulatory hypotheses.

7. Conclusion

7.1 Summary of Key Findings

This thesis set out to understand how the adoption of AI tools unfolds in cost engineering and procurement within an automotive OEM group, and to develop a structured basis for comparing and prioritizing tools across organizations at different stages of maturity and adoption. The answer required both a benchmarking instrument to compare tools across a common set of factors and recommendation logic to translate scores into adoption priorities. The central contribution is that these two analytical requirements can be satisfied within a single integrated framework, rather than through parallel and often conflicting exercises. What the study ultimately shows is that the challenge facing Company X and its peer organizations is not a technology deficit; the tools exist, the architectures are capable, and the use cases are understood. The challenge is an absorption deficit across workflow integration, data infrastructure, and ecosystem readiness that no single tool investment can resolve, but that a structured sequencing strategy, grounded in the benchmark scores, can address systematically.

The findings answered the three research questions as follows. RQ1 established that AI tools across the studied group cluster into two tiers: one directly serving cost estimation within the cost engineering workflow, and the other serving procurement support functions in the automotive OEM context, and that the cost estimation tools at peer organizations are further along in deployment than those at Company X, with supplier ecosystem readiness as the dominant industry-wide bottleneck. RQ2 established that the integrated TOE benchmark produces three diagnostic patterns: the pilot ceiling as a shared industry condition, the data-tool mismatch, and strategic anchoring as the dominant internal variable that a tool-by-tool ranking on its own would not reveal. RQ3 established that these ranked sequences can be translated into prioritized actions for Company X: engage with Company B on licensing for Tool B, conduct a commodity fit check for Tool D and adopt if needed, scale Tool XA now by performing sanity checks and protocol, maintain Tool XB without over-investing, monitor Tool C and Tool A for rollout and use case progress, respectively. Address the supplier ecosystem gap at the organization-wide strategy rather than at the tool-level and confirm every deployment as a named strategic decision. The consistency of this readiness picture across tool-level and function-level analysis strengthens confidence in the recommendations and confirms that the absorption deficit, not the technology gap, is the central challenge.

7.2 Contributions

The methodological contribution is the integrated TOE benchmark framework with six measurable factors, derived from interviews with experienced cost engineers at Company X, which anchors the framework in the practitioner perspective and in the practical need for AI-related tools. This benchmarking method allows the priority assessment to be performed in a single exercise for the case study company. This avoids the conceptual duplication of treating these as separate frameworks. The factor structure is anchored in literature (Appendix D), in

cost engineer and IT head experience (Appendix E), and in management requirements through the triangulation logic documented in Section 4.5. The reverse scoring logic on technical scope is a context-specific weighting that conventional maturity models, such as AIMM, AI-CAM, CMM, and CRISP-DM, cannot consider, and that this study's convergent evidence is based on the empirical statements of five out of six cost engineers and the IT head.

The empirical contribution is the tool-level evidence from Company X and four peer organizations, documented through seventeen semi-structured interviews, thirteen weeks of meeting observations, and a systematic review of forty-two AI-related papers. Most prior studies on AI related to cost engineering operate at the conceptual or functional level, and this study adds granular, practical insight at the tool-level, with explicit factor benchmarking across ten tools in a live industrial deployment context. The convergence between the tool-level benchmark scores and the functional-level assessment picture demonstrates that both levels of analysis are necessary for a robust adoption decision, and that they strengthen rather than replace each other when combined within the same analytical exercise.

The practical contribution is the prioritized recommendation logic for Company X, presented as the recommendation matrix in Table 8. The matrix translates benchmark scores into actionable priorities for the first-tier and second-tier tools and identifies the supplier ecosystem gap as a strategic-level concern requiring group-level coordination rather than tool-level investment. The same logic can be adapted by other industrial organizations facing similar AI selection and assessment decisions, especially within OEM group contexts where intra-group transferability of tools is high.

In closing, this study confirms that AI adoption in cost engineering is not a technology selection problem alone. It is an integrated problem of technology, organization, and environment that can only be resolved when all three dimensions are read together. The integrated TOE benchmark framework developed in this thesis offers Company X a practical instrument for navigating the problem and offers other industrial organizations a transferable approach for their own AI adoption journeys. Three conditions define what transferability requires: the framework must be re-triangulated to the specific operational context, the reverse scoring logic on Technical Scope must be assessed against the absorptive capacity of the receiving organization, and the benchmark must be treated as a point-in-time instrument that requires periodic re-scoring as tools, ecosystems, organizational capability, and readiness evolve in time. With those conditions met, the integrated TOE benchmark is not a one-time research instrument but a repeatable decision-support tool for AI adoption governance in the industrial cost engineering and procurement function.

8. Limitations and Future Research

8.1 Limitations of This Study

Five limitations limit the conclusions of this study. First, the scope is confined to one automotive OEM group. While the intra-group criteria strengthen analytical transferability based on the shared platforms, governance structures, and data standards, they also constrain statistical generalizability. The three cross-dimensional patterns identified in Chapter 5 are grounded in the studied group and should be treated as analytical propositions to be tested in other industrial contexts rather than as universal findings. Second, the benchmark is a cross-sectional instrument. Data collection took place over a defined period, and the level assignments reflect the maturity state of each tool at that point in time. Several tools were in active development during the study: Tool XA was mid-pilot, Tool XF was in the conceptual stage, and Tool B was in mid-rollout across additional brands. Longitudinal re-scoring would produce a different readiness picture and would allow adoption trajectories to be tracked over time rather than remaining a static inference.

Third, the reverse scoring logic on Technical Scope is a context-specific design choice grounded in triangulated evidence from the literature, from five of six cost engineers, and from the IT head interview. It is appropriate for the current capacity of Company X, but may not be appropriate in contexts where architectural complexity is an adoption enabler rather than a barrier. Researchers and practitioners applying the framework elsewhere should reassess this weighting against their own organizational context before adopting the scoring scheme unchanged. Fourth, the Ecosystem Readiness factor depends heavily on supplier-side data behavior, which the study can observe only from the OEM side. Supplier motivations for strategic data minimization, as identified by interviewee CE5, are not directly examined. A study incorporating the supplier perspective would produce a more complete reading of the ecosystem constraint and would test whether the bottleneck is primarily technical, commercial, or relational in origin. Fifth, the benchmark prioritizes tools serving the cost engineering function directly, which means the procurement use cases beyond cost estimation are not evaluated with the same depth. Organizations with a different functional priority may require additional factors or a different set of factors to apply to the framework.

8.2 Future Research Directions

Four directions for future research follow from the limitations of this study. First, a longitudinal re-scoring study tracking the same tools over an extended time would allow adoption trajectories to be mapped rather than only cross-sectional inferences to be drawn, and it would test whether the patterns, such as the pilot ceiling and the data-tool mismatch, represent transitional conditions or stable structural features. The integrated benchmark developed here is explicitly designed to support such a repeated application. Second, supplier-side research is needed. The Ecosystem Readiness constraint is observed exclusively from the OEM perspective, yet whether the bottleneck is primarily technical, commercial, or relational can only be resolved by incorporating tier-1 supplier voices into the empirical

dataset. Third, the cross-industry replication in sectors such as aerospace or construction equipment would test the framework's transferability and either validate or challenge the reverse scoring logic on Technical Scope in contexts with different capacity profiles. Fourth, integrating a user-level framework with the TOE benchmark would capture variability within the organization or team that can balance the scoring and would address the gap between the organization's deployments and what individual engineers use in practice.

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Appendices

Appendix A – Interview Guide One (Cost Engineers of Company X)

1. Can you briefly describe your role and your involvement in cost engineering or procurement, along with the overall cost estimation workflow in your organization?
2. Which parts of this workflow are manual versus system-supported, and where do you spend most of your time?
3. Which steps in the process are most time-consuming or repetitive, and where do you typically face errors or inefficiencies?
4. What challenges do you face regarding data (e.g., incomplete, inconsistent, or frequently changing data)?
5. Which activities in your workflow could be supported or automated using AI?
6. Have you seen similar AI tools used in other companies?
7. Which tasks would you trust AI to support in the cost estimation process?
8. What kind of cost insights would directly improve negotiation outcomes?
9. Would faster or automated cost estimation change your decisions?
10. What level of transparency or explainability do you expect from such tools?

Appendix B – Interview Guide Two (IT Head of Company X)

1. What AI or digital initiatives are currently active in procurement or engineering?
2. Are any of these related to cost estimation or decision support? What is their maturity (pilot or deployed)?
3. Are external partners or vendors involved in developing these solutions?
4. What is the roadmap for AI or digital development over the next 1–2 years?
5. What systems currently support cost engineering, and how well are they integrated?
6. Is data flow automated, or does it require manual handling between systems?
7. How would you assess data quality and availability for cost-related use cases?
8. Is data centralized or fragmented across systems?
9. What are the main technical barriers to introducing AI (e.g., integration, data format, infrastructure)?

10. How scalable is the current infrastructure for advanced analytics or AI?
11. How does IT collaborate with cost engineering and procurement today?
12. Are there dedicated resources (budget, teams) for AI or digital initiatives?
13. How are data ownership and governance structured?
14. What internal processes or approvals affect AI implementation?
15. What non-technical challenges could slow down or block AI adoption?
16. How does Company X's digital capability compare to competitors or peer organizations?
17. Are there group-level standards influencing AI or tool adoption?
18. What regulatory constraints (e.g., GDPR, data security) affect AI implementation?
19. Are there supplier or industry restrictions on data usage?
20. How do you ensure AI outputs are explainable and auditable?
21. Have compliance requirements ever delayed or limited AI initiatives?
22. From an IT perspective, where is AI adoption most feasible in cost engineering today?
23. Which areas are ready now, and which require further groundwork?
24. What key conditions must be in place to scale AI successfully?

Appendix C – Interview Guide Three (AI Tools Developers/Users)

1. What AI tools are currently used in cost engineering or procurement?
2. Which use cases deliver the most value in practice?
3. What types of technologies are used (e.g., AI, machine learning, rule-based systems, NLP, generative AI)?
4. Are these solutions standalone tools or integrated into existing systems (ERP, PLM, etc.)?
5. What is the maturity level of these solutions (concept, pilot, fully deployed)?
6. How widely are they used across the organization?
7. What measurable improvements have been observed (e.g., accuracy, cycle time, cost savings)?
8. How do these solutions compare to traditional methods?
9. What were the main challenges during implementation (technical or organizational)?
10. What changes were required in processes or workflows?

11. What external factors influenced AI adoption (e.g., competition, cost pressure, digital transformation strategy)?
12. Are there regulatory or compliance considerations affecting AI use?
13. How important is supplier data availability or ecosystem readiness?
14. What are the key success factors for implementing AI in cost engineering?
15. What should organizations avoid when adopting these solutions?
16. Which use cases would you recommend as a starting point for similar companies?

Appendix D – Thematic Analysis Coding of Literature Review (TOE Factors – Triangulation 1)

Factor	Sub-theme from literature	Description	Supporting papers (Author, Year)
Technical Scope	ML for cost estimation	Use of supervised ML, regression models and ANN as the core architecture for cost estimation in supply management.	Bodendorf, Merkl and Franke (2021); Hammann (2024); Mirasçı and Aksoy (2025); Manuguerra et al. (2023)
Technical Scope	Deep learning and NLP	Use of deep learning combined with activity-based costing, and NLP for supplier document similarity analysis.	Bodendorf and Franke (2024); Spreitzenbarth, Bode and Stuckenschmidt (2024); Bodendorf, Wytopil and Franke (2021)
Technical Scope	GenAI and LLM applications	Emerging use of LLMs and generative AI for procurement decision support and cost reasoning.	Mishra et al. (2024); Spreitzenbarth, Bode and Stuckenschmidt (2024)
Technical Scope	Hybrid and parametric methods	Combination of parametric models, outlier removal and ML for cost estimation.	Öfele (2021); Sessa (2020)
Use Cases & Value	Should-costing and price prediction	Use of AI for should-costing of new parts, price prediction and quotation support.	Hammann (2024); Mirasçı and Aksoy (2025); Mandolini et al. (2024); Bodendorf, Merkl and Franke (2021)
Use Cases & Value	Supplier evaluation and selection	AI for supplier selection, total cost of ownership prediction and supplier scorecards.	Spreitzenbarth and Struckenschmidt (2025); O'Connor, Romero and Asiaei (2026); Lange (2025)

Use Cases & Value	Spend analytics and contract analysis	AI for spend analysis, contract review and procurement decision support.	Balkan and Akyuz (2025); Spreitzenbarth (2024); Bodendorf, Wytopil and Franke (2021)
Use Cases & Value	Process automation and RPA	Use of RPA and process automation for procurement workflow improvement.	Flechsig et al. (2022); Althabatah et al. (2023)
Excluded	Explainability / Interpretability	The need for AI results to be auditable and defensible in supplier negotiations. Embedded within the Workflow & Culture factor as black-box skepticism, but not as a stand-alone factor.	Bodendorf and Franke (2024); Mishra et al. (2024)
Excluded	Technology Acceptance	Application of TAM to intelligent cost estimation systems. Absorbed into Workflow & Culture as user adoption resistance.	Bodendorf and Franke (2022); Riti, Bacali and Abrudan (2025)
Excluded	Engineering Change Management	Data-based methods for adaptive engineering change systems. Adjacent to cost engineering but not directly within the factor structure.	Schuh et al. (2021)

Appendix E – Thematic Analysis Coding of Interview Study One and Two (TOE Factors – Triangulation 2)

Factor	Sub theme	Description	Codes	Quotes	Interviewee	Level
Technical Scope	GenAI deployed in daily workflow	Active personal use of ChatGPT and Claude for cost breakdown analysis; the only cost engineer in the dataset operating at this level.	ChatGPT, Claude, GenAI in daily use, cost breakdown summarization, individual deployment	90-95% automized for me AI (ChatGPT/Claude) is taking over general information analysis. Whatever I receive as cost breakdown Excel, I upload to AI and navigate it to summaries.	CE3	L3
	Autonomous costing capacity	technical scope is framed as an aspirational autonomous costing capability, where AI could	Autonomous costing, AI-led calculation, cost-calculation automation, expert-task	Because AI can do all the calculations with what we are doing as an experiment.	CE1	L4

		perform the full cost-calculation task currently carried out by the cost engineer, while still requiring final checking due to possible errors.	automation, human proofreading	They might also give some error. they need a proof reading of the final Sometimes we are doing the estimation so they can do		
Integration	Future fully integrated AI vision	Aspirational state where AI reads RFQ, takes information, populates internal cost estimation tool, and performs calculation autonomously.	Fully integrated AI, automatic RFQ reading, autonomous internal cost estimation tool population, end-to-end automation	If you will build... when a purchaser sent the RFQ. It may read all the documents and take the information and then give it... do the calculations. If we will fully integrate, so they don't need us even for purchasers.	CE1	L4
	Tool A standalone with manual transfer	Tool A operates as a standalone calculation environment; data transfer to and from other systems is manual.	Tool A standalone, manual data transfer, no API connection, separate committee sheet	Tool A is a standalone tool. We need to manually transfer information. The AI assistant is integrated [in Teams] but not to Tool A.	CE4	L1
	Standalone tools, manual calculations	Mechanical tools used independently with manual calculation transfer; no integrated workflow.	Standalone tools, manual calculation, no integration, separate analysis steps	All standalone, all manual. Each calculation step is done in isolation.	CE5	L1
Use case	Quantified RFQ cycle time reduction	Concrete time-saving target: a significant reduction in cycle time for RFQ analysis if AI replaces manual market analysis.	Significant cycle time reduction for RFQs, AI target, market analysis automation, electronic parts commodity scope	A significant cycle time reduction for RFQs is expected with AI. Market analysis for electronic parts gives part numbers and distributor links.	CE2	L2
	Predictive spend analytics	High-value future use case: link to raw	Predictive spend analytics, raw	Predictive spend analytics link to raw material	CE6	L3

		material market and update database prices in real time.	material market linkage, real-time pricing, dynamic database	market, update database prices in real time.		
	AI use case for price estimation	Primary identified procurement use case: predict cost of a part before negotiation with suppliers.	Price estimation, part cost prediction, pre-negotiation, identified but not executed	AI use case for price estimation's ability to predict cost of a part before negotiation with suppliers. There hasn't been any execution on that.	ITH	L3
	Negotiation Readiness	AI-supported costing can strengthen negotiation readiness by preparing calculation outputs that the cost engineer can use directly in supplier discussions, provided the outputs are transparent and validated through sanity checks.	Negotiation readiness, AI-supported costing, sanity check, cost argument preparation, supplier negotiation support	It would be good if it's it provides everything OK and you just do the sanity check. OK, makes sense. I understand. I can go into a negotiation with that numbers. Makes sense. And then, yeah, this would help a lot, yeah.	CE4	L4
Workflow	Change management as primary org barrier	Explicit framing of the cultural challenge: long Tool A tenure makes the team mindset slow to shift.	Change management, mindset, Tool A long tenure, AI teaching effort	Change management is the mindset. People used to Tool A for very long time. Teaching AI the way the product is looked this will take a lot of energy.	CE3	L2
	Black-box AI skepticism	Strong organizational fear of black-box AI; transparency is described as the	Black-box skepticism, fear of AI, transparency prerequisite, trust building	Black-box skepticism people afraid of AI. It's really a cost calculation the	CE4	L1

		prerequisite for trust.		split and the machines behind this would be helpful.		
External Drivers	Strategic digital transformation	External pressure framed as strategic necessity rather than active urgency; transformation as imperative.	Digital transformation, strategy, in the end we must do that, time saving	Digital transformation, we can say. Yeah, it's more like a strategy. In the end we must do that. Time saving also works because sometimes you might get many cost breakdown at the same time.	CE1	L1
	Regulatory constraints on AI use	Awareness of regulatory restrictions; concrete example of US activity halts due to compliance.	Regulatory extremely important, USA stopped working with AI, compliance constraints	Especially USA stopped to work with it because of regulatory is extremely important. We're going to receive a lot of issues, so it is the main problem today.	CE1	L2
	Industry-wide AI race awareness	Strong sense that the industry is in active AI experimentation, but no single solution has emerged as a winner.	Industry-wide AI experimentation, no winning solution, race posture	Everyone is trying but no one has the solution yet.	CE3	L4
	Company X as early AI enterprise adopter	External validation of Company X's strategic position: recognized externally as an early enterprise adopter of AI.	Early AI adopter, enterprise license, external recognition, doing well for automotive	Company X is one of the early adopters of an AI enterprise license. So, in terms of adoption, considering that we are an automotive company and not like Spotify or a tech savvy company, I think we're doing well.	ITH	L4

Ecosystem Readiness	Shared data ownership, ecosystem governance	shared governance of supplier/customer data, where data ownership is not controlled by one party alone but becomes usable across the AI-costing ecosystem under agreed rules.	Shared data ownership, ecosystem governance, supplier–customer data sharing, AI data access, collaborative AI-costing	there should be some mutual consideration between two parties like supplier, customer that it is going to be available for the third party The owner of data needs to be changed. It shouldn't be some. It shouldn't be. It's one-party thing	CE3	L4
	Supplier as structured information sharing	ecosystem readiness is reflected through structured supplier interaction, where suppliers provide technical presentations to explain their concept and support cost engineers in understanding the product/process logic.	Supplier presentation, structured supplier input, concept explanation, supplier technical information, ecosystem interaction	So, you ask for a supplier presentation, and they present the concept and then you get an idea list	CE4	L3
	Multiple business flows data inconsistency	Two different business flows create data inconsistency that any AI tool inherits.	Multiple business flows, system switching, data inconsistency	In Company X we have the industrial flow and all the procurement that we do to be able to build a heavy vehicle in the manufacturing process. And after we sell the vehicle... that is another part of the business that handles that. So, there could be some inconsistencies.	ITH	L2

Trust calibration with AI output	Sanity checks as institutional design	Cost engineers consistently described a need for designated human roles to validate AI output before it enters negotiation.	Designated checker role, human verification, transparency requirement, calibrated trust	Should not trust AI results as it is.	CE2	-
Generational and skill divide	Prompting skill heterogeneity	AI productivity varies widely across staff; some have prompting skill, most do not; younger staff tend to enter with more AI exposure.	Prompting skill varies widely, generational gap, university AI exposure, upskilling agenda, change management	Not everybody has prompting skill. How to lift the level of competence is something we are also discussing.	ITH	-

Appendix F - Thematic Analysis Coding of Interview Study Three (Tool Benchmark and Level Assignment Table)

Tool A (Company A)

Factor	Quote	Keywords	Level	Score
Technical Scope	Maybe this AI assistant can fit into, I think Gen. AI. Yeah, for the moment there's no connection [to Tool A], so we use it just for questions, summarizing. It's a software product we're getting from Microsoft.	Generative AI (general AI assistant), Microsoft product, no Tool A -in active AI, summarizing only	L2	3
Integration	Connected to our software of calculation. Basically, that is the PLM platform. right now, we don't have a lot of connections. We don't use any specific AI for cost engineering in the moment, just standardized things like the AI assistant chatting with it in the end.	Standalone AI assistant, no Tool A connection, separate productivity tool	L1	1
Use Cases & Value	It's more to prepare things or ask quick questions but not related directly to costing. We don't use any specific AI for cost engineering now.	Quick data gathering, generic productivity, no measurable value for cost calculation	L1	1

Workflow & Culture	We started, I would say last year. So maybe we have less than one year experience with the AI assistant. We were not responsible for implementing it, it's not a specific tool just for cost engineering. The IT department provided it. Change management is here a critical point, especially when you have the gap of younger employees who already have much experience using AI out of university.	Recently introduced, IT-led implementation, change management gap, generational divide	L2	2
External Drivers	The pressure from outside. I mean, you must use it to be able to be competitive in the future. In my opinion, the chatbot is just the beginning.	Competitive pressure, market necessity, beginning of a journey	L1	1
Ecosystem Readiness	For sure there is something, but we are not responsible. Our compliance in purchasing they for sure must ensure that it's not getting sensible information. So just sign something that it's secure.	No supplier-side data integration considered, compliance-only stance, low data-safety maturity	L1	1

Tool B (Company B)

Factor	Quote	Keywords	Level	Score
Technical Scope	We're using a fuzzy matching algorithm where you can auto-link some stuff. Yes, exactly, we are currently working on using an LLM to be able to match... different cost breakdowns together automatically.	Fuzzy matching algorithm, rule-based today, LLM under development, structured comparison	L1	4
Integration	It's live at Company B, but we are still developing it. So, it's not everything is fleshed out and we are developing new features. We're currently working with Company C to also get an interface with Legacy systems. So, all the new cost breakdowns get also automatically uploaded in the tool, but this is still something that we are working on.	Live in production, ongoing feature development, planned upstream interface to GPS, restricted user roles	L2	2
Use Cases & Value	Where we are trying to estimate part price of a new part using the information of similar or old parts. So the AI is actually used to read the drawings, read the 3D models, 2D drawings... and come back with a similar part and then go into the cost breakdowns and give us a calculation of the cost should-costing of a new part. We go over all of the cost breakdowns, we take a look at, for example, everything that is a	Should-costing of new parts, similar-part matching, raw material market linkage, structured cost breakdown comparison, multi-use case	L3	3

	raw material and then we classify it and match it to a current market price.			
Workflow & Culture	We have to respect data restrictions that's why series buyers have restrictions in their parts groups where project buyers, cost engineers or managers can see everything. We expect to roll it out to the other business units later this year. Every brand that uses the tool has to pay a licensing fee because we cannot give it for free for legal/tax reasons.	Standard workflow with role-based restrictions, planned multi-unit rollout, established licensing/governance	L3	3
External Drivers	The reason why is he wants to connect all the initiatives that we have regarding cost breakdown in the group so we can align and don't do any double work.	Centralized cost breakdown initiative, alignment strategy, group-level coordination, anti-duplication directive	L3	3
Ecosystem Readiness	It's also not yet over the whole database. We have to do a lot of data quality checks and I'm still validating if there are issues. If you want to use cost breakdown data, I think you will always run into the issue of data quality either you would have to be very strict and only say we want to use cost breakdown that we have validated or if you want to use all of the database then you will always have these kind of issues.	Manual data uploads, data quality challenges, no supplier-direct integration, validation effort acknowledged	L1	1

Tool C — (Company C)

Factor	Quote	Keywords	Level	Score
Technical Scope	This tool is just using an LLM. This LLM was tuned just to understand the requirements in the spec books. We have a huge database based upon already calculated stuff... based upon this database we trained our LLM.	LLM-based, fine-tuned on departmental data, NLP for spec interpretation, advanced single-modality AI	L3	2
Integration	We are right now in this special phase where we try just to implement some stuff. We have already the sizing tool created, but we are not able to deploy it so far. Maybe in one or two weeks. We need some special regulation from our company. IT needs to look on it. Then we need approval from the board.	Deployment pending, IT governance review, board approval required, not yet integrated to production systems	L1	1

Use Cases & Value	What we are already doing, we are sizing the spec books based on AI. We're using it to size the spec book this is one of the biggest use cases. Usually it just takes us several weeks to calculate or, if you have many parts, then for sure you need a lot of time and knowledge to evaluate it. But these tools can do it within an hour or less, so it's a huge improvement... almost 80% of the whole workflow steps optimized.	Spec book sizing, requirements analysis, significant time reduction, major workflow optimization	L2	2
Workflow & Culture	We have a kind of a hidden IT team here in value engineering. That's the biggest point. If you go to the IT department to get this, you need to approve it, you need the business case... that's why we implement two guys working really embedded. They're supporting, they know the business, they understand it, they have also this direct connection. They can include all this knowledge in a very fast way without a lot of coordination.	Embedded IT team in value engineering, business-IT proximity, change initiated, dedicated AI advocates	L2	2
External Drivers	As a company we need to be fast. Everybody's talking about China speed, you need to be the first in the market or at least not the last one. The mindset of the customer is changing. If you need to fulfil, they need to love you. That's the main goal. Just a car maker we are more, we are providing mobility... If you are not using the current state-of-the-art technologies, then maybe you need to change your company and go to the vintage companies.	Cost-to-company pressure, customer satisfaction, strategy change, market positioning, technology necessity	L3	3
Ecosystem Readiness	Data is one of the biggest, it's like gold. Well, data is everything. If you just look at what kind of data is used right now, so they are already limited. If you include all this data, then you need new data. So new data will be maybe also created by AI. So, it has the same quality, maybe... So, data from humans, direct customers, that's what we will get, will be very important.	Data limitations acknowledged, internal data only, no supplier integration, future human-generated data envisioned	L1	1

Tool D (Company D)

Factor	Quote	Keywords	Level	Score
Technical Scope	What we are using is the machine learning, supervised learning, computer vision. We make an image or more images from one PCB and then we will have the opportunity to get the costs on every component.	Machine learning, computer vision, supervised learning, image-to-Bill of Materials, single-purpose AI architecture	L2	3
Integration	It's a standalone tool. It was specially developed for us from us we have a machine learning team and for the front end and back end we get some help of an external partner.	Standalone tool, in-house ML team, manual picture upload, external partner for user interface	L1	1
Use Cases & Value	In 2019 I had the idea that if we have a PCBA, we don't have to count manually all the components. We can use in the benchmarking. For example, competitors currently you can do all the benchmark with our application. You have the claim management and commodities. So that means if we have a claim, we will get the whole Electronic Unit, we analyze and then we will see what we have for now in our project and what was nominated.	PCB cost prediction from images, library matching, claim management, competitive benchmarking, multi-use case in single tool	L3	3
Workflow & Culture	To get all the approval from the IT guys, that was one of the hardest points in our organization. To be honest, it was very easy to build it technically. It's especially Tool D, so we are in a roll-out at other peer organization, for example. This peer organization does not work on AI development tool but gets the application, and Trading like Company X and Company A is also interested.	IT approval as hardest barrier, deployment achieved, multi-unit rollout, organizational adaptation	L2	2
External Drivers	To use AI in our organization, especially in procurement, means that you don't need so much human resources. Our departments will be more efficient with the AI solutions.	Efficiency pressure, headcount efficiency, cost-to-company driver, strategic resource optimization	L3	3
Ecosystem Readiness	To be honest, we as the Company C group, I'm not sure that we will have such partnerships with our suppliers where we can share all the information we need and also, they need. Our organization didn't know how powerful partnership could be in such an environment with our suppliers so on. I think you have more partnership in the international OEMs.	No supplier partnership, ecosystem-readiness gap acknowledged, group-level cultural barrier, peer benchmark with international OEMs	L1	1

Tool XA (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	It's like learning by doing in an LLM. It's also giving the user information on how you should use this data. It's just not a number. It's going to show you why this number is here. Tool XA must think like a buyer or a procurement person.	LLM-based, learning-by-doing, contextual reasoning, buyer-like behavior, advanced GenAI architecture	L3	2
Integration	It's only searching for specific rows and specific data and extracting that and modifying the data and presenting it in a better way. It doesn't feed anything back. It's not a standalone or so it's not integrated. So, it's in between.	Manual data input, no feedback loop to source systems, between-states integration, extraction-only	L2	2
Use Cases & Value	Cost estimation simple stamping parts. Last week we started to upload files. Before then you provided them with information regarding a limited number of parts. Explaining to Tool XA development team, how we look at numbers and how we use them the reasoning.	Cost estimation, simple stamping parts, limited pilot scope, reasoning-based delivery	L2	2
Workflow & Culture	Pilot. Pilot. Pilot. Yeah. We are starting with the internal teams. We have our first demo for them next week. So, it's not used. I cannot say how much have been into it.	Pilot stage, change initiated, specific team to lead, individual champions emerging, demo scheduled	L2	2
External Drivers	Yes, but one cost everything trickles down to cost. Efficiency equals cost and digital transformation equals cost. So, I think the whole digitalization. From an owner-type point of view, the owner will say yes, we save time, we save money. This is cost driven internally for Company X	Driving digital transformation, cost volatility, internal cost-driven mandate, group-aligned strategy	L3	3
Ecosystem Readiness	It's not like Tool XA development team is heavily working on supplier ecosystem. They are working on it but not as heavily as they should. Or maybe it's not so important to them they focus on the production line, the physical part.	Supplier ecosystem not considered, vendor focus on production data, cost engineering ecosystem secondary	L1	1

Tool XB (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	I use Tool XB and that is not an AI at all. It's very simple mathematical logic. It's some kind of advanced Excel basically that I can do this coding in the background to build the models. It's very simple coding. I can just copy-paste these mathematical formulas and put in the code.	Rule-based, parametric/mathematical logic, advanced Excel-like, no AI, custom user-built models	L1	4
Integration	I only have a limited number of licenses. You know, it works a a substantial number of people here, but not all of them are CAD designers. Currently the maturity level of all these software's is almost like the pilots, but it can be improved further.	Limited license pool, partial deployment, designer-only access, pilot-level integration	L1	1
Use Cases & Value	We put it in the hands of designers, they can do their own costing, and by doing that we will get lower product cost and shorter lead times. We can get answers much faster by simulating on our own. We will most likely get the lower cost out of it.	Designer-driven costing, lower product cost, shorter lead times, decentralized cost analysis	L2	2
Workflow & Culture	It's not their primary task to do cost analysis, so it means that they will not work with this like every day; it will be maybe one time a week or one time a month, and that's why the software must be so simple.	Non-primary task adoption, low-frequency use, simplicity requirement, limited adoption	L2	2
External Drivers	Competition for sure. So that it also means we must be even faster in our development and then we're looking at AI tools to just be more efficient. It's just a competitive that we need to be more efficient as an organization. Less waste and just work faster.	Competitive pressure, R&D efficiency, faster development, less waste	L1	1
Ecosystem Readiness	Most of our own CAD models, I mean, that's our own design, so we have the best possibility to do our own costing analysis. If it's not available, then it's not available for a good reason. Maybe you can use AI to make the analysis anyway, but probably you need a cost engineer to do it.	Internal CAD data only, no supplier integration, manual analysis fallback when CAD unavailable	L1	1

Tool XC (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	We are now trying to develop this supplier risk profile with Tool XA development team, with the aim to have basically one dashboard that will enable us to visualize all the risks that might affect our supply chain. There is a lot of data mining through, of course, the internal systems, and the whole dashboard is done in rule-based algorithms.	Multiple AI mix rule-based with data mining, GenAI/ML evolving, pattern recognition, automated alerts	L2	3
Integration	It's a web app today. It has a connection and is receiving information from the different systems that we have. 14 different internal Company X systems. Will connect to third-party providers because part of the data we don't have today the capabilities for identifying certain types of risks.	Web-based dashboard, 14+ legacy system connections, third-party integration planned, pilot-level live deployment	L2	2
Use Cases & Value	Visualize all the risks that might affect our supply chain risk different risk dimensions, from natural hazards and natural disasters to geopolitical structural risks. Top 10 risk suppliers, what are the risks. Deep dive in each risk dimension to understand the performance. This is an activity that consumes at least 12 hours a month for the finance department. Right now, with Tool XA development team and you can just click with one button, and it will immediately show you which supplier has been degraded. Once you identify the supplier degraded, you can click one more button and it will shoot an email automatically to the purchaser.	Multi-dimensional risk visibility, automated financial monitoring, 12-hour-per-month effort eliminated, automated alerting	L3	3
Workflow & Culture	We are a pilot now. Information is scattered across different systems, and they don't speak to each other. We are lacking so much information today and spending so much time in data gathering.	Pilot-stage replacing manual workflow, fragmented current state, time-saving narrative driving adoption	L2	2

External Drivers	Cost pressure is one of those. We know that whenever we are not efficient, we end up paying that at a much higher price because we cannot stop the line. So, we end up paying our freights. We end up paying 40% extra margin to get apart from a distributor. But also it's part of the digital transformation strategy that we have.	Cost pressure, line-stop avoidance, margin protection, digital transformation strategy	L1	1
Ecosystem Readiness	We depend on external business intelligence. The quality of the data is essential. If suppliers are not providing the master data, then we lack data.	Strong dependency on internal and external data quality, supplier master-data dependence, BI-vendor reliance	L2	2

Tool XD (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	What it does is whatever we do manually, I do that in an Excel, but what it does is it goes to different Power Business Intelligence, it goes to Legacy system, it goes to spend report.	Multi-source GenAI integration, rule-based tracking, real-time updates, automation of Excel workflow	L2	3
Integration	It's not a standalone, it is integrated with the systems that we already have.	Integrated with legacy systems, end-to-end pipeline, final-stage deployment	L3	3
Use Cases & Value	This app will help the buyers to track the payment, to see what's the current payment terms and how, where their target is, and how to track the payment terms when they have negotiated with the buyers, is that updated in the system. So, this Tool XD will be end to end.	Payment term tracking, monitoring, end-to-end supplier financial performance, buyer-facing dashboard	L2	2
Workflow & Culture	Right now, we do everything in an Excel. It's still done; we put them in our internal collaboration platform every month and then it's trackable, so it's already widely used. But now we will only change how we will track it.	Excel-based manual workflow, internal collaboration platform tracking, change initiated to integrated tool, widely used today	L2	2
External Drivers	If Company X has lower payment terms for this middle-sized company, then we will go up in media saying that Company X is not paying or has bigger payment terms for suppliers who are small-sized.	Regulatory pressure, governance, media risk, supplier fairness, payment-term compliance	L2	2

Ecosystem Readiness	No, no, we don't need their input. We already have a supplier data in our systems; we just use that.	Internal data only, no supplier integration required, but inconsistent internal data sources	L2	2

Tool XE (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	We have created an agent which you can ask. The idea is that you don't have to know what you're searching for. For example, you don't need to know the number of the instruction, but you can ask, for example, how do I introduce a supplier?	Generative AI agent, natural language query interpretation, document retrieval, NLP-based intent matching	L2	3
Integration	It's integrated in Teams. So, everybody will be able to see it.	Integrated within Microsoft Teams, accessible to all users via existing collaboration platform	L2	2
Use Cases & Value	It's about either you find it in there with the agent, or you don't find it. But you're still saving time and it's about we have results from 20 to 30 seconds on finding documents.	Intelligent document retrieval, time saving in document search (20-30 seconds), binary success/failure outcome	L1	1
Workflow & Culture	It's somewhere between pilot and fully deployed. We only have, it's about 40 people that has tested it. I think it's not we're quite manual still, I would say. But it's helping more as an assistant.	Pilot phase, 40-person test group, gradual rollout planned, assistant positioning, underlying work still manual	L2	2
External Drivers	To make us more efficient and more modern.	Digital transformation push, efficiency improvement, internal modernization	L3	3
Ecosystem Readiness	Maybe not to our AI assistant. It's important to have, and maybe more towards cost engineering here. But towards my agent, it's not important.	Internally driven knowledge system, supplier ecosystem not relevant to this tool, internal-only data	L1	1

Tool XF (Company X)

Factor	Quote	Keywords	Level	Score
Technical Scope	What we want, we want to have a toll XF. We want to be able not only to take what we have in the Power Business Intelligence reports and have the historical data. We also would like to have the outlook, what's happening around the world. I would say the most area is generative AI.	Generative AI supported, historical + forward-looking analysis, predictive insights, pattern identification	L3	2
Integration	We don't want you to have a different application. We want you then to be integrated in the tools that we use.	Integrated application across Power Business Intelligence, legacy systems, invoices, logistics, contracts; embedded in existing tools	L2	2
Use Cases & Value	It's about gathering the data. It's about having the tools because today we spend 98% of the time to gathering the data and 2% of the time doing the actual work that we should be doing. Information is power and the more we can empower the purchasers, the better the outcome we can achieve.	Reduced data-gathering effort (98% less), decision-empowerment, value reframing of buyer time	L1	1
Workflow & Culture	We are not mature enough as an organization to do that. At least a change of mindset is needed.	Low organizational maturity, mindset-change requirement, pilot-stage adoption	L2	2
External Drivers	It's external factors. For me it's about survival if we are going to have, based on the world we live in today, an industry in Europe. With how it looks to compete with China, for example, we need to be good on this, otherwise we won't have an industry.	Competitive pressure, survival imperative, China competition, European industrial positioning	L1	1
Ecosystem Readiness	Data, data is everything. The integration between customer and supplier is not in the same way that we have today, where you have a clear interface between us and them.	Future supplier-customer integration envisioned, data trust weak, supplier interface immature	L1	1